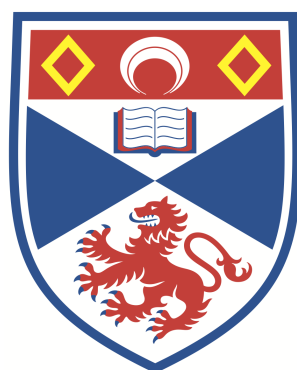


Bayesian Gaussian Process Hyperparameter Inference for Implied Volatility Surfaces

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Abstract

Many financial quantitative risk valuation methods rely on accurate interpolation of implied volatility surfaces, yet traditional non-parametric models often struggle to provide this precision due to the microstructure noise inherent to financial data. While Gaussian Process regression offers a highly flexible modelling solution, standard deterministic point estimation techniques are highly susceptible to this noise contamination, resulting in weakly identifiable log-marginal likelihood landscapes. In response, the model overfits the microstructural market frictions (such as bid-ask spreads) by learning the surface and artificially inflating the OOS RMSE indicator. The result is a jagged implied volatility surface that violates strict static arbitrage constraints, including calendar and butterfly conditions.

As a solution to the instability of the LML topology, this dissertation introduces a fully Bayesian framework utilising the Hamiltonian Monte Carlo (HMC) No-U-Turn Sampler (NUTS) to integrate over hyperparameter uncertainty rather than optimising for a single coordinate. The primary innovation of this research lies in framing Bayesian hyperparameter marginalisation as a structural regulariser, naturally filtering out the high-frequency overfitting that causes arbitrage violations in the first place. This approach fundamentally prevents the emergence of economic arbitrage violations without the necessity to artificially restrict the surface.

The empirical evaluations in S&P 500 index options across diverse macroeconomic regimes demonstrate that the deterministic optimiser suffers a 61.9% butterfly arbitrage failure rate on evaluated dates, frequently collapsing into systemic dual-violations. In contrast, the fully Bayesian framework acts as a structural regulariser by marginalising over hyperparameter uncertainty bounded by the SVI-inspired prior. The Hamiltonian Monte Carlo sampler acts solely as the computational engine to navigate this posterior space, allowing the model to drastically reduce, and often eradicate, these localised arbitrage violations. As a trade-off, we have to accept a microscopic penalty to the out-of-sample generalisation error in low-volatility regimes. This proves to be a mandatory and mathematically acceptable trade-off to guarantee an economically admissible, arbitrage-free volatility surface.

Declaration

I declare that this dissertation is my own work and that it has not been submitted, in whole or in part, for any other degree or diploma. All sources used have been acknowledged appropriately.

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I would like to thank my supervisor for his guidance and the experience he has provided during the development of this research. Additionally, I would like to extend my gratitude to my family and friends for their unconditional support throughout this period.

Contents

Abstract	i
Declaration	ii
Acknowledgements	iii
Glossary of Financial Terminology	1
I. Basic Option Mechanics	1
II. Volatility & Surface Metrics	3
III. Arbitrage & Pricing Theory	5
1 Introduction	7
1.1 Context and Motivation	7
1.2 Problem Statement	7
1.3 Research Questions and Hypothesis	8
1.4 Dissertation Structure	9
2 Financial Foundations and Constraints	10
2.1 The Implied Volatility Surface	10
2.2 Static Arbitrage Constraints	11
2.2.1 Calendar Arbitrage	12
2.2.2 Butterfly Arbitrage	13
2.3 The Parametric Prior	14
3 Gaussian Processes in Financial Modelling	16
3.1 GP Regression Framework	16
3.2 The Matérn 5/2 Kernel	18
3.3 The Standard Approach: Point Estimation	20
4 The Pathology of Weak Identifiability	23
4.1 Mapping the LML Topology	23
4.2 Restart Instability	24
5 Bayesian Hyperparameter Inference via HMC	26

5.1	From Optimisation to Integration	26
5.2	Markov Chain Monte Carlo	27
5.3	Hamiltonian Monte Carlo	28
5.3.1	Phase Space and the Hamiltonian	29
5.3.2	Hamiltonian Dynamics	29
5.3.3	Theoretical Guarantees: Conservation, Volume, and Reversibility	30
5.3.4	The No-U-Turn Sampler (NUTS)	32
6	Empirical Methodology and Experimental Design	36
6.1	Data Ingestion and Preprocessing	36
6.2	Market Regimes	37
6.3	Evaluation Metrics	38
7	Empirical Results and Diagnostics	40
7.1	Predictive Accuracy: The Global Fit	40
7.2	Visualising the Posterior Diagnostics	41
7.3	Financial Consistency and the Arbitrage Trade-off	43
7.4	The Accuracy-Consistency Trade-off	45
7.4.1	The Illusion of Deterministic Superiority	45
7.4.2	Empirical Case Studies: The 2022 and 2023 Regimes	46
8	Conclusion	47
8.1	The Pathology of Point Estimation	47
8.2	The Accuracy-Consistency Trade-off	47
8.3	Computational Limitations and Future Work	48
A	Pipeline Architecture	50
B	Mathematical Proofs and Extended Derivations	51
B.1	The Schur Complement and the GP Predictive Distribution	51
B.1.1	The Setup: The Joint Multivariate Gaussian	51
B.1.2	The Derivation: Affine Transformation and the Schur Complement	52
B.1.3	Decorrelation and Independence	53
B.1.4	The Result: The Conditional Predictive Distribution	53
B.2	Matrix Calculus of the Log-Marginal Likelihood Gradients	54
B.2.1	The Setup: The Objective Function	54
B.2.2	Mathematical Prerequisites: Matrix Calculus Rules	54
B.2.3	The Derivation: Differentiating $\mathcal{L}(\theta)$	55
B.2.4	The Result: The Gradient Equation	56
B.3	Analytical Derivatives of the Matérn 5/2 Kernel	57

B.3.1	The GP Linear Operator Property	57
B.3.2	The Matérn 5/2 Kernel and Mahalanobis Distance	57
B.3.3	The Derivation: First Derivative (The Gradient)	58
B.3.4	The Derivation: Second Derivative (The Hessian)	59
B.4	The Breeden-Litzenberger Density and Static Arbitrage	60
B.4.1	The Setup: Risk-Neutral Expected Payoff	60
B.4.2	The Derivation: First Derivative (The Survival Function)	60
B.4.3	The Derivation: Second Derivative (The Density)	61
B.4.4	The Result: The Butterfly Arbitrage Constraint	61
B.5	Dupire's Equation	62
B.5.1	The Setup: The SDE and the FPK Equation	62
B.5.2	The Derivation: Differentiating the Call Price with Respect to Time	62
B.5.3	Substituting the FPK Equation	63
B.5.4	Connecting to Breeden-Litzenberger	64
B.5.5	The Result: Dupire's Formula	65
B.6	Mercer's Theorem and the RKHS Norm Derivation	66
B.6.1	Definition of the Kernel Operator	66
B.6.2	Mercer's Theorem (Eigen-decomposition)	66
B.6.3	Construction of the RKHS ($\mathcal{H}_k$)	67
B.6.4	Verification of the Reproducing Property	67
B.6.5	Derivation of the RKHS Norm	67
B.6.6	Application to the Gaussian Process Loss Functional	68
B.7	Fisher Information Matrix (FIM) for a Gaussian Process	69
B.7.1	The Log-Marginal Likelihood	69
B.7.2	The Score Function	69
B.7.3	The Second Derivative (The Hessian)	69
B.7.4	The Expectation	70
B.7.5	The Result: Fisher Information Matrix	70

C Comprehensive Empirical Results 72

List of Figures

1	A canonical representation of an Implied Volatility Surface, demonstrating the volatility skew across log-moneyness and maturity.	3
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4.1	Topology on 3 March 2020.	23
7.1	LML topology on 23 March 2023.	41
7.2	Posterior distributions of the Matérn length-scales during the High Volatility Shock (3 March 2020). The sampler achieves proper mixing, comprehensively exploring the weakly identifiable ridge rather than collapsing into localized boundary modes.	42
7.3	Posterior distributions of the Matérn length-scales during the High Momentum Expansion (15 February 2024). The sampler became more confident as the true economic shape was more defined.	43
7.4	3D implied volatility surface comparison on 2024-01-24. The deterministic L-BFGS-B model overfits local noise into arbitrage-violating non-convexities, whereas full Bayesian HMC marginalisation guarantees a smooth, structurally consistent predictive surface.	44

List of Tables

7.1	Out-of-Sample Predictive Accuracy during the March 2020 High Volatility Regime.	41
7.2	Aggregate Arbitrage Failure Rates across the 21 unique historical trading days.	45
7.3	The Accuracy-Consistency Trade-off, where C. Viol. and B. Viol. represent the calendar and butterfly arbitrage violations respectively. The Point-Estimated GP achieves a slightly lower RMSE by overfitting microstructural noise, generating dozens of arbitrage violations.	46
C.1	Master Comparison of Out-of-Sample RMSE and Arbitrage Violations: Point-Estimated GP vs. Bayesian HMC GP.	73

Glossary of Financial Terminology

This is a comprehensive glossary with fundamental financial terminology used throughout this dissertation.

I. Basic Option Definitions

These terms define the fundamental building blocks of the derivative contracts modelled in this text.

Option Contract: A financial derivative that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specific price on or before a certain date.

European-style Option: An option contract that can only be exercised at the expiration date.

American Option: In contrast to the European Option, an American option contract allows the holder to exercise their right at any time prior to expiration.

Call Option: A contract whose payoff is tied to the right to buy the underlying asset, defined mathematically at maturity as $(S_T - K)^+ = \max(S_T - K, 0)$. Intuitively, an investor might buy a call to speculate if they anticipate asset appreciation; if that happens, they will be able to purchase the asset at the Strike Price K .

Strike Price (K): The predetermined, fixed price at which the option contract can be executed.

Spot Price (S_0): Current or initial market price.

Maturity / Time to Maturity (T): The annualised time remaining until the option contract expires.

Premium: The absolute market price of the option today, reflecting the compensation traders demand for bearing risk.

SPX: The ticker symbol for S&P 500 Index options, which track the largest Fortune 500 companies and serve as the highly liquid (i. e., an asset or market allows fast buying or selling at stable prices without changing the asset's value), primary benchmark for global equity derivative markets.

Bid-Ask Spread / Microstructural Noise: The small-scale market frictions and discrete tick sizes present in real-world trading that introduce noise into observed option prices.

II. Volatility & Surface Metrics

These definitions bridge the gap between absolute option prices and the standardised space where the Gaussian Process operates.

Black-Scholes Model: The Black and Scholes [13] model can be used to price European-style options and to estimate a theoretical fair value for a call or put based on spot, strike, time to expiry, risk-free rate, and volatility. It is also widely used as a benchmark for hedging and for *inferring implied volatility* from market option prices.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

where, $V = V(S, t)$ is the option value, S is the underlying asset price, t is time, r is the risk-free interest rate and σ is the volatility of the underlying asset returns.

Black-Scholes Implied Volatility (σ_{imp}): The unique volatility parameter that, when passed through the inverted Black-Scholes-Merton formula, yields the exact market price of the option. It acts as a forward-looking metric of market consensus on risk.

Implied Volatility Surface: A three-dimensional graph mapping implied volatility across various strikes and maturities for a single asset, denoted as $(K, T) \mapsto \sigma_{imp}(K, T)$. An example can be:

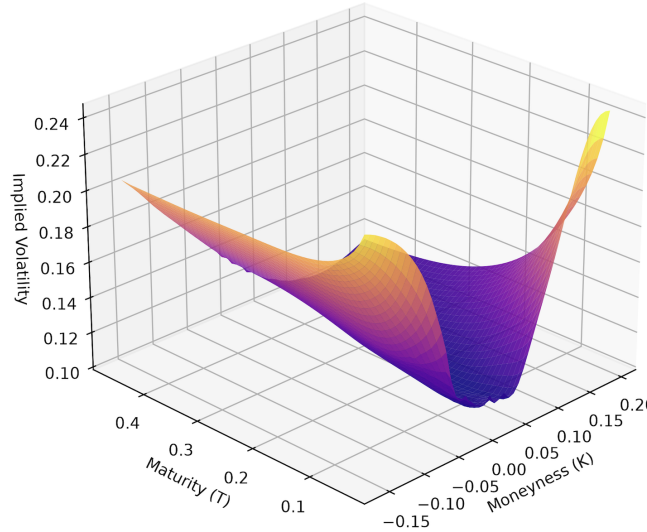


Figure 1: A canonical representation of an Implied Volatility Surface, demonstrating the volatility skew across log-moneyness and maturity.

Log-Moneyness (k): A normalised metric defined as $k = \log(K/F_0)$ used to gauge risk independent of the underlying asset's price dynamics, where F_0 denotes the asset's forward price at time zero.

Forward Price (F_0): The agreed-upon price of the underlying asset for future delivery at time zero, used to calculate log-moneyness.

Total Variance (w): A transformation of the implied volatility to total variance space using the equation $w(k, T) = \sigma_{imp}^2(k, T)T$. First, it is easier to calculate arbitrage violations in total variance space because variance should be non-decreasing across strikes and maturities. In addition, $\sigma^2 T$ is much easier to implement into SDE's than $\sigma\sqrt{T}$.

Volatility Skew / Smile: The structural, non-linear shape of the volatility surface across strikes, often exhibiting extreme downside asymmetry in equity markets like the SPX. Can often be referred to as a skewed convex parabola for simplicity.

Out-of-the-money (OTM): Option contracts with strikes far from the current asset price; deep OTM options encode the extreme asymptotic tails of the market's risk-neutral distribution.

In-the-money (ITM) and At-the-money (ATM): Option contracts with strike equal to the Spot Price (ATM) and option contracts with strike less than the Spot Price (ITM).

III. Arbitrage & Pricing Theory

These concepts define the strict financial and mathematical boundaries that the model must respect to remain valid.

Risk-Neutral Measure ($\mathbb{Q}$): A central asset pricing framework under which the expected return of every asset equals the risk-free rate; this means that we can calculate the price of an asset using expected values. Effectively behaving like martingales, which was first shown by Harrison and Kreps [9].

Risk-Free Rate (r): The continuously compounded annual rate of return of an investment with zero risk. For example, an initial investment of \$100 at a continuously compounded rate of 5% for 1 year. The future value is $100e^{0.05 \cdot \frac{12}{12}} \approx 105.127\$$.

Yield Curve: A yield curve or an interest rate graph plots the cost of borrowing at the time to maturity. A parallel shift in the yield curve implies a raise or a reduction in both short- and long-term rates, where the shape of the curve does not change.

Static Arbitrage: A financial scenario where a model implies the existence of a portfolio that has a non-negative future payoff but costs less than zero today, allowing for risk-free profit. Financially, this implies a self-financing portfolio with a strictly positive probability of a riskless profit.

Calendar Spread: An option strategy involving two contracts with the same strike price but different maturities. For example, a trader might sell a near-term 1-month call option at strike $K = 5000$ and simultaneously buy a longer-term 3-month call option at the exact same strike $K = 5000$ to profit from time-decay or term structure shifts.

Butterfly Spread: A neutral options strategy involving three staggered strike prices ($K_1 < K_2 < K_3$). Typically, a trader buys one lower strike option at K_1 , sells two middle strike options at K_2 , and buys one higher strike option at K_3 . In the context of volatility surfaces, pricing a butterfly spread below zero indicates a negative risk-neutral probability, signifying an arbitrage violation.

Calendar Arbitrage: A violation occurring when a longer-maturity option has a lower total variance than a shorter-maturity one. Essentially, an arbitrage opportunity arises when a longer-dated option is priced lower than its shorter-dated counterpart. This violates the rule that uncertainty must accumulate monotonically over time ($\frac{\partial w}{\partial T} \geq 0$).

Butterfly Arbitrage: A violation occurring when a butterfly spread is priced below zero, implying a negative risk-neutral probability density. This manifests as a breakdown

of convexity in the call price curve across strikes ($\frac{\partial^2 C}{\partial K^2} < 0$), yielding an economically inadmissible volatility surface.

Breeden-Litzenberger Theorem: A fundamental financial theorem that connects local concavity of a financial surface to butterfly arbitrage violations. ($\frac{\partial^2 C}{\partial K^2}$).

Constrained Kriging: A spatial interpolation method in Gaussian Process regression where hard constraints (such as monotonicity or convexity) are explicitly enforced on the predictive posterior to satisfy financial arbitrage conditions, often at the expense of computational efficiency or model flexibility.

Backwardation: A market regime characterized by an inverted term structure. In the context of implied volatility surfaces, backwardation occurs when near-term expected volatility is strictly greater than long-term expected volatility, usually signaling acute, immediate macroeconomic distress.

Call Skew: An asymmetric structural deformation in the volatility smile where out-of-the-money (OTM) call options command higher implied volatilities than equidistant OTM put options. This typically emerges during aggressive, high-momentum market expansions.

Exotic Options (Asian / Barrier): A broad class of derivative contracts possessing payoffs that are strictly path-dependent, contrasting with standard European options whose payoffs are determined solely by the terminal asset state S_T .

Chapter 1

Introduction

This chapter introduces the motivation and outlines the scope of the research the paper will address. It also defines the central hypothesis and the structure of the dissertation. The formal definitions and context for the problem will begin in Chapter 2.

1.1 Context and Motivation

Accurate modelling of the implied volatility surface is central to modern quantitative finance. Volatility surfaces are required for pricing, hedging, calibration, and risk management, particularly in index option markets such as S&P 500 Index (SPX). At the same time, Bayesian nonparametric methods, and Gaussian Processes in particular, offer a flexible framework for modelling nonlinear structure with uncertainty quantification.

This dissertation is motivated by the need for a reliable implied-volatility surface for pricing and risk management. Because financial data inherently contains microstructural noise, Gaussian Process regression is a flexible yet reliable solution that has the potential to interpolate this sparse and noisy volatility data smoothly. However, the practical performance of the regression is heavily dependent on the estimation of unstable kernel hyperparameters. This paper studies whether the Bayesian approach to those hyperparameters gives a more reliable surface.

1.2 Problem Statement

The problem considered in this dissertation is the failure of the deterministic point estimation algorithm in weakly identifiable topological landscapes of the Log-Marginal Likelihood for Gaussian Process volatility surface models. Under noisy and chaotic market data, the L-BFGS-B algorithm suffers from arbitrary hyperparameter configurations, depending on the initialisation coordinate. Because the ridge at the bottom of the LML surface is flat, any point yields near-equivalent log-marginal likelihood values but results in different predictive surfaces and pricing outputs.

1.3 Research Questions and Hypothesis

We can break up the objective into two main research questions:

- How does the weak identifiability of Gaussian Process hyperparameters impact the out-of-sample predictive accuracy of the implied volatility surface?
- Can Bayesian inference via Hamiltonian Monte Carlo (HMC) and the No-U-Turn Sampler (NUTS) [10, 19] resolve this instability to improve pricing performance while maintaining arbitrage-free constraints?

In this dissertation, we adopt the central hypothesis that flat ridges in the log-marginal likelihood topology cause deterministic optimisers to fail, and that replacing point estimation with a full posterior distribution over hyperparameters will enforce structural regularisation, drastically reducing localized butterfly and calendar arbitrage violations, with only a minimal and acceptable trade-off in nominal out-of-sample prediction accuracy.

Although the individual mathematical and financial components of this hypothesis have been explored separately in recent literature, their synthesis represents a novel contribution to the modelling of implied volatility. Within the topic of machine learning, the pathology of weakly identifiable Log-Marginal Likelihood (LML) topologies is well-documented and researched; papers like Titsias [3] have specifically outlined the shortcomings of the point estimation method. Hamiltonian Monte Carlo (HMC) is frequently proposed as an elegant solution to resolving the overfitting inherent in the deterministic optimiser [19].

Additionally, Gaussian Processes have been explored for volatility surface interpolation in recent quantitative finance literature. Cousin & Gueye [2] and Chataigner et al. [1] have recognised that GP is a powerful non-parametric tool for surface interpolation. However, existing finance literature has often artificially specified strict arbitrage violations by relying on constrained kriging (see [III. Arbitrage & Pricing Theory](#)) to explicitly avoid calendar and butterfly arbitrage violations. This thesis, on the other hand, seeks to bridge these two independent domains by establishing a direct causal link. Rather than treating arbitrage violation as a purely financial constraint that we need to artificially enforce, this research identifies the LML flat-ridge pathology and the resulting memorisation of microstructural noise as the fundamental mathematical cause of these economic violations.

To summarise, the novel contribution of this work lies in framing Bayesian hyperparameter marginalisation as a structural regulariser, naturally filtering out the high-frequency overfitting that causes arbitrage violations in the first place.

1.4 Dissertation Structure

The dissertation is organised into eight chapters. Chapter 2 introduces the financial and arbitrage foundations of the implied volatility surface. Chapter 3 develops the Gaussian Process framework and point-estimation approach. Chapter 4 empirically identifies the weak-identifiability pathology. Chapter 5 introduces Bayesian hyperparameter inference using HMC/NUTS. Chapter 6 describes the data and empirical design. Chapter 7 presents the results, and Chapter 8 concludes with implications, limitations, and future work.

Chapter 2

Financial Foundations and Constraints

2.1 The Implied Volatility Surface

Before discussing the financial constraints, we should introduce the object to which they will apply: the implied volatility surface. The surface is a three-dimensional graph that maps the volatility of option contracts across several maturities and strikes (see [I. Basic Option Mechanics](#)) for a single asset. Specifically, a function mapping of an option's strike K and time to maturity T can be defined as $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$ across available strikes and maturities. The annualised volatility figure is derived from the inverted Black-Scholes-Merton formula (see [II. Volatility & Surface Metrics](#)) to standardise absolute option prices. This standardisation enables comparison of an option across different strikes and maturities.

Importantly, the Black-Scholes implied volatility (denoted as $\sigma_{\text{imp}}(K, T)$) (see [II. Volatility & Surface Metrics](#)) is the unique volatility parameter that, when used in the Black and Scholes [13] model, yields the option's exact market price. It acts as a forward-looking metric, pricing in the market consensus on expected risk and uncertainty rather than past price movements. In practice, it reflects the premium or compensation that traders demand today for bearing risk. This is because longer-dated options inherently carry greater variance, meaning the asset's price has more time to rise above the strike, which poses additional risk for the option issuer.

This expected risk is difficult to evaluate by only looking at absolute metrics such as spot (see [I. Basic Option Mechanics](#)) and option prices. Hence, we should take a more relative approach in order to gauge this uncertainty, which implies constructing log-moneyness, defined as $k = \log(K/F_0)$ (see [II. Volatility & Surface Metrics](#)). By using logarithms, we produce a stable and normalised representation of the volatility surface, independent of the underlying asset's price dynamics.

The Role of SPX Options. For practical purposes, the underlying asset for this empirical evaluation is the S&P 500 Index (SPX), which tracks the performance of

the largest U.S. Fortune 500 companies. The SPX derivatives serve as the primary benchmark for global equity derivative markets. This is due to a couple of reasons: the SPX being exceptionally liquid (i. e., an asset or market allows fast buying or selling at stable prices without changing the asset's value) across many strikes and maturities, possessing a European-style contract type (see [I. Basic Option Mechanics](#)) and abundant documentation. Additionally, the SPX surface possesses a highly pronounced structural downside skew. This extreme asymmetry makes SPX options a highly interesting testing ground for evaluating Bayesian hyperparameter inference and surface modelling.

2.2 Static Arbitrage Constraints

Before introducing the Gaussian Process model, we must specify the constraints that will ensure the implied volatility surface is not only mathematically, but also financially and legally sound. The two static arbitrage constraints that are of most importance for this purpose are the *calendar arbitrage* and the *butterfly arbitrage* (see [III. Arbitrage & Pricing Theory](#)). These constraints allow for the surface to have a valid risk-neutral probability density function. In turn, this makes it financially impossible if it implies the existence of a portfolio with a non-negative payoff and a negative cost. Hence, a mathematically smooth and stable interpolant is not sufficient; it must also be economically admissible.

Another important topic of discussion that formalises the aforementioned is a framework central to asset pricing: the risk-neutral principle (see [III. Arbitrage & Pricing Theory](#)). Intuitively, we use the risk-neutral measure $\mathbb{Q}$ in derivatives pricing as an equivalent probability measure to the physical measure $\mathbb{P}$, defined via the Radon–Nikodym derivative density $\frac{d\mathbb{Q}}{d\mathbb{P}}$ [9, 20]. It reweights the real-world probability distribution by shifting the expected asset return to the risk-free rate r (see [III. Arbitrage & Pricing Theory](#)). This parameter transformation absorbs investor risk aversion into the probability density itself. Because this framework relies on evaluating densities, the measure $\mathbb{Q}$ must strictly satisfy the foundational axioms of probability theory proposed by Kolmogorov [25], allowing us to compute no-arbitrage derivative prices as rigorous discounted expectations $\mathbb{E}^{\mathbb{Q}}[e^{-rT} X_T]$.

Proposition 2.1 (Arbitrage-free call price under risk-neutral measure $\mathbb{Q}$). *Strictly, if X_T is a payoff at maturity T and the term S represents the current spot price of the underlying asset, then its time-0 value is*

$$X_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[X_T],$$

where r is the continuously compounded risk-free rate and $\mathbb{Q}$ is the risk-neutral measure. For a European call option with strike K , the payoff is

$$(S_T - K)^+ = \max(S_T - K, 0),$$

so its arbitrage-free price is

$$C(K, T) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

This pricing identity is the starting point for both static-arbitrage conditions: calendar arbitrage is linked to the option price rising over time, while butterfly arbitrage is linked to the payoff profile having a non-negative probability density.

However, these conditions are difficult to implement through pure implied volatility (σ_{BS}), which makes static arbitrage constraints highly non-linear and mathematically intractable. A paper on Stochastic Volatility-Inspired (SVI) Surfaces by Gatheral and Jacquier [8] pioneered the idea of shifting to total variance space $w(k, \tau)$, which linearises the no-arbitrage conditions, making them far easier to enforce mathematically. The relationship of implied volatility to total variance is straightforward:

$$w(k, T) = \sigma_{\text{imp}}^2(k, T) T$$

where $k = \log(K/S_0)$ is log-moneyness. We prefer the total variance space for two reasons. Firstly, it removes the trivial $\sqrt{T}$ scaling present in implied volatility, thereby making IV surfaces across different maturities comparable on a uniform scale. The proof is trivial and omitted for brevity. Secondly, the calendar arbitrage conditions can be naturally expressed as monotonicity conditions on $w(k, T)$, while butterfly arbitrage conditions can be checked after converting the fitted surface back into call prices.

In other words, the total variance space serves as a smooth framework that allows us to conveniently assert the financial constraints, which we will discuss further.

2.2.1 Calendar Arbitrage

As described by Gatheral and Jacquier [8], calendar arbitrage arises from the calendar spread option strategy, which involves two options with the same strike price but different maturities. An arbitrage opportunity arises when the longer-maturity option has a lower total variance than the shorter-maturity one. This makes the near-month T_1 contract overpriced, which in turn could be shorted (i. e., taking a negative position in the asset to capture the price discrepancy), and the far-month T_2 contract underpriced ($T_1 < T_2$). It goes without saying that this applies only to European options or non-dividend-paying American options, which we will assume from now on. In practical terms, the total variance should increase monotonically with respect to time maturity T , a foundational conclusion proved by Gatheral [7].

Link to total variance. In total variance terms, the calendar arbitrage constraint becomes a monotonicity condition in maturity:

$$\frac{\partial w(k, T)}{\partial T} \geq 0.$$

The $w(k, T)$ represents the cumulative variance up to time T , the constraint implies that with every step, the variance should not decrease. In other words, the uncertainty should accumulate over time and never decrease; otherwise, it would create an arbitrage opportunity.

Proposition 2.2 (Calendar monotonicity). *For each fixed strike K , the map $T \mapsto C(K, T)$ is nondecreasing. Consequently, the total implied variance*

$$w(k, T) = \sigma_{\text{imp}}^2(k, T) T$$

is nondecreasing in T for each fixed log-moneyness k ; in particular, if w is differentiable,

$$\frac{\partial w}{\partial T}(k, T) \geq 0.$$

Proof. Let $0 < T_1 < T_2$. No-arbitrage implies $C(K, T_2) \geq C(K, T_1)$. Since the Black–Scholes call price is strictly increasing in total variance, the result is as follows. $\square$

2.2.2 Butterfly Arbitrage

Butterfly arbitrage originates from the butterfly spread option strategy, which involves four options with staggered strike prices $K_3 > K_2 > K_1$. It entails purchasing a call at a low strike price K_1 , selling two calls at a middle strike price K_2 , and buying a call at a high strike price K_3 (where $K_3 - K_2 = K_2 - K_1$). This arbitrage indicates that the implied probability distribution of the asset is negative and invalid, often due to an overly steep or improperly smoothed IV surface during interpolation. Consequently, the position's payoff can become negative, meaning the trader profits risk-free without owning the underlying asset.

Intuition. A natural way to test this constraint is to examine the local curvature; the curve should always be convex, so the middle strike should not lie above the average of the neighbouring strikes. In other words, if the middle option price is too high relative to the neighbouring points, an arbitrage opportunity exists. The connection to the fundamental Breeden–Litzenberger theorem can be found in Appendix B.4, where the proof is that a sufficiently tight butterfly spread is a finite-difference estimate of the second derivative of the call price curve with respect to strike K . It is important to note that in the case of a butterfly arbitrage violation (implying probabilities), the local volatility in Dupire's

framework becomes mathematically undefined, we direct the reader to Appendix B.5 for the full breakdown.

Relationship to total variance. As we proved, butterfly arbitrage can be naturally stated in terms of call prices, unlike calendar arbitrage, which is inherently a monotonicity condition on the total variance across maturities. However, since the fitted model is expressed in the total-variance space, the implied total variance space must be converted back to call prices before assessing the convexity condition. Simply, this is done using the Black–Scholes formula:

$$C_{\text{BS}}(S, K, T, r, \sigma) = S N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

In the simplified Black–Scholes formula (see II. Volatility & Surface Metrics), C_{BS} denotes the theoretical fair value of a European call option. The term S represents the current spot price of the underlying asset, while K is the strike price, the predetermined price threshold at which the contract can be exercised. The annualized time remaining until contract expiration is denoted by T , and r represents the continuously compounded risk-free interest rate used in the present-value discount factor e^{-rT} . The parameter σ specifies volatility, defined as the annualized standard deviation of logarithmic asset returns. For statistical context, $N(\cdot)$ is the standard normal cumulative distribution function $\Phi(\cdot)$, where $N(d_2) = \mathbb{Q}(S_T > K)$ represents the risk-neutral probability that the terminal asset price exceeds the strike price (finishing "in-the-money"), and $N(d_1)$ corresponds to the option's delta, which weights the expected asset value received upon exercise.

2.3 The Parametric Prior

In Bayesian nonparametric machine learning models like Gaussian Processes, setting a mean function lets us include financial knowledge before seeing any data. Rather than making the Gaussian Process learn the overall shape of the implied volatility surface from nothing, we use a parametric prior. We define the prior distribution as:

$$w(K, T) = T \left(|a| + |b|k^2 + ck \right). \quad (2.1)$$

The mean function uses a quadratic polynomial in log-moneyness, which is heavily based on the Stochastic Volatility-Inspired (SVI) parametrisation (Gatheral and Jacquier

[8]). By avoiding the full five-parameter SVI, which is difficult to optimise, the resulting model is essentially a parabolic curve in log-moneyness. Where k^2 (i. e., log-moneyness) provides the characteristic convex smile, k is the volatility skew, and T scales it across maturities.

Originally, the standard Gatheral's raw SVI formula is indifferent to the definition of variance, which can never be negative. Hence, the inclusion of absolute values for a and b . Alternatively, a standard unconstrained optimiser (e.g., Nelder-Mead or L-BFGS-B) might blindly test for negative values, causing the total variance to become negative and rendering the pricing algorithm mathematically undefined. By incorporating $|a|$ and $|b|$, a differentiable barrier is created where the optimizer can search anywhere from $-\infty$ to ∞ freely while mathematically guaranteeing a positive total variance.

Finally, this setup also works well with the tail behaviour suggested by the moment-based constraints model. In other words, deep OTM (see [II. Volatility & Surface Metrics](#)) options encode the tails of the risk-neutral distribution; their prices must remain consistent with that distribution. The prior is not just smoothing the data, but it also encodes a theoretically allowed tail growth rate for the implied variance, based on which moments of the underlying asset are finite. The far OTM smile should match the asymptotic limits set by Roger Lee's moment formula bounds [11]. Even though there is little real-option data far out of the money, this structured prior keeps the extrapolated total variance within economic bounds and follows the theoretical maximum growth rates of the volatility smile at extreme strikes.

Chapter 3

Gaussian Processes in Financial Modelling

3.1 GP Regression Framework

We formally define a Gaussian Process (GP) as a collection of random variables, any finite subset of which possesses a joint Gaussian distribution. To ensure the mathematical existence of this stochastic process across an uncountable index set, we rely on the foundational probability space $(\Omega, \mathcal{F}, \mathbb{P})$ established by Kolmogorov [25].

Consequently, this construction provides a strictly non-parametric Bayesian framework for performing inference over continuous function spaces. A GP is completely specified by its mean function $m(\mathbf{x})$ and its covariance function, or kernel, $k(\mathbf{x}, \mathbf{x}')$. We denote the Gaussian Process prior as:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (3.1)$$

The mean function, defined as $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$, gives the GP the global topology of the function space. Conceptually, it encodes the prior distribution or an assumption about the distribution of an unknown function. It is set to dictate the overall smoothness, behaviour and interpolation of the unknown function.

In standard machine learning literature, the mean function $m(\mathbf{x})$ is often set to zero for algebraic convenience; however, a financial surface is inherently structurally asymmetrical. This asymmetry can be observed in features such as the persistent equity skew. Hence, the mean function needs careful tuning in order to safely extrapolate quotes in the sparsely populated deep out-of-the-money tails. To address this, the framework explicitly defines $m(\mathbf{x})$ using the parametric prior formulated in Equation 2.1:

$$m(\mathbf{x}) = T(|a| + |b|k^2 + ck)$$

By substituting this SVI-inspired formulation into the GP mean function, the global

structural constraints and theoretical asymptotic tail bounds are embedded directly in the prior distribution. Consequently, the learning burden shifts: the GP is no longer required to infer the macroeconomic shape of the smile from raw data. Instead, it is solely tasked with modelling the residuals. The residuals represent the localised, non-parametric deviations of the observed market data from the parametric mean. These residuals are governed by the covariance kernel, $k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))]$. Essentially, this is the covariance between the process values at $\mathbf{x}$ and $\mathbf{x}'$, and $k(\mathbf{x}, \mathbf{x}')$ measures how these residuals vary together. The kernel describes how points on the volatility surface are related. It controls how smooth the surface is and how quickly the effect of one data point fades as we move across strike and term.

To formalise the regression model, assume a dataset of n noisy total variance observations $\mathbf{y} = \{y_1, \dots, y_n\}^\top$ evaluated at input locations $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The observation model is given by:

$$y_i = f(\mathbf{x}_i) + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$ represents independent, identically distributed Gaussian noise. In finance, the noise variance term (σ_n^2) is essential for accounting for market microstructure noise, discrete tick sizes, and the effects of the bid-ask spread.

Given a set of test inputs X_* , the joint distribution of the training targets $\mathbf{y}$ and the predictive function values $\mathbf{f}_*$ under the GP prior is multivariate Gaussian:

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(X) \\ m(X_*) \end{bmatrix}, \begin{bmatrix} K(X, X) + \sigma_n^2 I & K(X, X_*) \\ K(X_*, X) & K(X_*, X_*) \end{bmatrix} \right) \quad (3.2)$$

where $K(X, X)$ is the $n \times n$ Gram matrix evaluated on the training inputs, $K(X, X_*)$ is the $n \times n_*$ cross-covariance matrix between the training X and test X_* inputs, and I is the identity matrix. The statement 3.2 asserts that by combining the observed training inputs and unknown test inputs together, the resulting joint distribution is Gaussian under the GP model. Importantly, the left block is the covariance between observed training inputs, plus the noise $\sigma^2 I$ term, which indicates that the observed training targets $\mathbf{y}$ are noisy versions of the underlying function values. In this case, the model allows for noise rather than treating the observed data as exact realisations of the latent function.

By conditioning this joint prior on the observed market data $\mathbf{y}$, we derive the predictive posterior distribution $\mathbf{f}_* | X, \mathbf{y}, X_* \sim \mathcal{N}(\boldsymbol{\mu}_*, \Sigma_*)$ as shown in the B.1 derivation. Equivalently, the above expression states that once we know the training outputs, we can update the joint Gaussian to get a new Gaussian distribution for the unknown test outputs. This is a standard conditioning step in GP regression. The exact matrix equations for the predictive mean and predictive covariance are:

$$\boldsymbol{\mu}_* = m(X_*) + K(X_*, X)[K(X, X) + \sigma_n^2 I]^{-1}(\mathbf{y} - m(X))$$

The predictive mean equation states that the forecast at the new points is the sum of the prior mean and a correction term, which is the covariance between the test and training inputs, weighted by the inverse of the covariance between training inputs, multiplied by the GP deviations from the mean function.

$$\boldsymbol{\Sigma}_* = K(X_*, X_*) - K(X_*, X)[K(X, X) + \sigma_n^2 I]^{-1}K(X, X_*)$$

The predictive covariance formula provides the model with the uncertainty after observing the data, where $K(X_*, X_*)$ is the prior uncertainty at the test points minus the part explained by the training data. In practical terms, uncertainty is minimised in dense regions and remains higher where the model must extrapolate.

3.2 The Matérn 5/2 Kernel

While the mean function determines the overall shape of the volatility surface, the covariance kernel governs the local shape at discrete points and the interpolative behaviour of the residuals. The choice of kernel cannot be arbitrary; in financial applications, a covariance function should satisfy strict structural requirements dictated by arbitrage pricing theory. In this work, we adopt an anisotropic Matérn 5/2 kernel; other common kernels (RBF, Matérn 3/2, Rational Quadratic) are viable in general-purpose modelling but have features that are discussed briefly below, which make them less suitable for the present arbitrage-constrained framework.

Because the extraction of a valid risk-neutral probability density via the Breeden-Litzenberger theorem is based on the evaluation of the second partial derivative with respect to strike K , a Matérn 3/2 cannot be chosen as it produces sample paths that are only once mean-square differentiable. Implying that strict convexity must be enforced directly. On the other hand, the Squared Exponential (RBF) kernel generates infinitely differentiable, analytic sample paths, imposing an excessive degree of smoothing that fails to capture the localized volatility spikes and microstructural frictions inherent in equity markets. Hence, the choice leans toward the twice mean-square differentiable ($\mathcal{C}^2$) Matérn 5/2 kernel, which satisfies the theoretical prerequisites of arbitrage pricing theory while retaining sufficient flexibility to model complex residual geometries.

The covariance kernel is a function of two inputs, $\mathbf{x}$ and $\mathbf{x}'$. We define two distinct input vectors: $\mathbf{x} = (k, T)^\top$ and $\mathbf{x}' = (k', T')^\top$. Furthermore, to account for the fundamentally different dynamics along the strike and term-structure dimensions, we define the anisotropic Mahalanobis distance r using independent length-scales ℓ_k and ℓ_T :

$$r(\mathbf{x}, \mathbf{x}') = \sqrt{\frac{(k - k')^2}{\ell_k^2} + \frac{(T - T')^2}{\ell_T^2}}$$

The Matérn 5/2 covariance function is then formally defined as:

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right) \exp(-\sqrt{5}r)$$

where σ_f^2 represents the signal variance, controlling the vertical amplitude of the latent Gaussian Process.

Justification and Differentiability Constraints The Matérn class of covariance functions is characterised by its versatility and smoothness parameter ν , which allows for explicit control of the underlying random field. This is especially useful for Implied Volatility surfaces, as they require a twice-differentiable kernel to extract arbitrage-free risk-neutral probabilities from the surface. As established in standard Gaussian Process literature by Rasmussen and Williams [12], the sample paths of a GP with a Matérn kernel are m times mean-square differentiable if and only if $\nu > m$. Consequently, setting $\nu = 5/2$ in $k_{5/2}(r)$ guarantees that the resulting latent function space is exactly twice mean-square differentiable ($\mathcal{C}^2$). The continuous calculus required to prove the kernel supports static arbitrage bounds is provided in Appendix B.3.

This exact $\mathcal{C}^2$ regularity is almost a requirement when modelling implied volatility surfaces. The popular Squared Exponential (or Gaussian) kernel, which is the case when ν approaches infinity, gives sample paths that are infinitely smooth and analytic. However, this level of smoothness does not match real financial markets, since it over-smooths local volatility spikes and misses the small-scale frictions seen in options trading. The resultant surface is highly analytic, which facilitates stable extrapolation for far OTM options, but pricing in between quotes is too restrictive. On the other hand, if $\nu = 3/2$ or $\nu = 1/2$, the resulting functions are only once differentiable or not differentiable at all, as in the Ornstein-Uhlenbeck process [5].

As mentioned before, the reason for exactly twice-differentiable sample paths is linked to the extraction of the market-implied risk-neutral density. As derived from the Breeden-Litzenberger theorem [4], the risk-neutral probability density $p(K, T)$ is strictly proportional to the second partial derivative of the call price with respect to the strike:

$$p(K, T) = e^{rT} \frac{\partial^2 C}{\partial K^2}$$

The total variance surface $w(k, T)$, mapped by the GP, needs to be inverted using the Black-Scholes formula to calculate call prices. The Black-Scholes model needs a volatility input to produce call prices. Once we know $w(k, T)$, we convert it to IV and price the option. For this to be an arbitrage-free, the underlying interpolant must allow

stable second-order differentiation. On the other hand, if the sample paths are not twice differentiable, the finite-difference methods used for computing $\frac{\partial^2 C}{\partial K^2}$ can become unstable, undefined or show singularities, which could falsely indicate butterfly arbitrage violations.

In brief, the Matérn kernel with $\nu = 5/2$ specification offers an ideal balance between smoothness and restriction. It creates smooth curves while still giving the algorithm enough flexibility to accommodate localised peaks and valleys. Additionally, because ν is a fraction, it involves simpler polynomial and exponential terms that reduce cost complexity and prevent mathematical instability.

3.3 The Standard Approach: Point Estimation

The standard paradigm for training a Gaussian Process in applied machine learning is to select the hyperparameters θ that maximise the probability of the observed data [12]. This is achieved by maximising the log-marginal likelihood (LML). Conditioning the joint Gaussian prior on the observed target vector $\mathbf{y}$, we define the log-marginal likelihood as:

$$\log p(\mathbf{y}|X, \theta) = -\frac{1}{2}\mathbf{y}^\top K_y^{-1}\mathbf{y} - \frac{1}{2}\log |K_y| - \frac{n}{2}\log(2\pi) \quad (3.3)$$

where

$$K_y = K_\theta + \sigma_n^2 I \quad (3.4)$$

represents the noisy covariance matrix.

To understand why hyperparameter optimisation often fails in the financial setting, we need to deconstruct the LML function. The kernel hyperparameters are estimated by maximising the Logarithm of Marginal Likelihood (LML), which balances data fit and model complexity. In practical terms, the LML balances model flexibility against data interpolation. This can be seen by decomposing the formula into three distinct theoretical components:

$$\log p(\mathbf{y}|X, \theta) = \underbrace{-\frac{1}{2}\mathbf{y}^\top K_y^{-1}\mathbf{y}}_{\text{Data Fit}} - \underbrace{\frac{1}{2}\log |K_y|}_{\text{Complexity Penalty}} - \underbrace{\frac{n}{2}\log(2\pi)}_{\text{Normalization Constant}}$$

The fundamental geometry of LML is a balance between the first two terms:

1. **The Data Fit (Mahalanobis Distance):** The term $-\frac{1}{2}\mathbf{y}^\top K_y^{-1}\mathbf{y}$ measures how well the observed outputs align with the covariance structure, calculated via the Mahalanobis distance between the observed data and the GP prior mean (which is zeroed out here for the residuals). Additionally, K_y^{-1} reweights the data based on what points should co-vary, not just when the residuals are small but also when those residuals fit the prior's structure. This term strongly prefers highly complex, flexible models. If we allow the Matérn length-scales (ℓ_K and ℓ_T) to shrink toward zero, the

covariance between distinct points vanishes, K_y approaches a scaled identity matrix, and the GP essentially memorises the training data. Consequently, the model loses all of its extrapolation capabilities. A model that perfectly interpolates every noisy observation will drive this data fit penalty toward zero, pushing the LML higher.

2. **The Complexity Penalty (Log-Determinant):** The term $-\frac{1}{2} \log |K_y|$ acts as an inherent regularisation mechanism. When the length-scales are small, the points one unit apart become weakly correlated. Hence, the kernel matrix is nearly diagonal, changing the log-determinant term. In other words, when the model allows the function to vary widely from point to point, the penalty becomes more negative, reducing the marginal likelihood. This regularisation mechanism serves as the formal RKHS norm constraint preventing jagged surfaces, Appendix B.6 formally defines the norm using functional analysis.

The Origin of Weak Identifiability (Flat Ridges): In an ideal scenario, the competition between the Data Fit and the Complexity penalty results in a convex, unimodal peak in the LML landscape. In this case, the algorithm can easily identify a single optimal parameter set $\hat{\theta}$. On the other hand, financial data exhibit characteristic noise and friction (e.g., bid-ask spread noise). Where the LML may exhibit weak identifiability when different hyperparameter combinations yield nearly the same covariance matrix K_θ . This is because the option market data can be successfully interpolated using multiple combinations of strike and maturity length-scales. Mathematically, at a local optimum $\hat{\theta}$, the Hessian of the negative LML measures local curvature. If the Hessian has a very small eigenvalue, then the LML changes slowly in the corresponding direction. Geometrically, this produces a plateau or flat ridges with weakly identifiable hyperparameters.

Standard Software Implementation and the Mathematical Failure of quasi-Newton Optimisation: Standard Gaussian Process libraries (such as `scikit-learn` [18]) navigate this objective landscape using L-BFGS-B (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with Bounds) [22], a deterministic quasi-Newton optimisation algorithm.

Let $\mathcal{L}(\theta) = \log p(\mathbf{y}|X, \theta)$. L-BFGS-B seeks to maximise the LML by iteratively solving a local quadratic approximation centred at the current hyperparameter estimate θ_k :

$$\mathcal{L}(\theta_k + p) \approx \mathcal{L}(\theta_k) + \nabla \mathcal{L}(\theta_k)^\top p + \frac{1}{2} p^\top \mathcal{H}_k p,$$

where $\mathcal{H}_k = \nabla^2 \mathcal{L}(\theta_k)$ is the Hessian. A second-order Taylor expansion forms the local quadratic model underlying quasi-Newton methods [16], as the LML is a complicated nonlinear function of θ with no closed-form optimal step.

However, $\mathcal{H}_k$ is computationally prohibitive to compute exactly. This is due to the extensive differentiation sequences of the kernel matrix, which involve terms of the form

$\text{tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_i} \right)$. These are expensive when N is large. Moreover, the Hessian can become nearly singular, for instance, when hyperparameters are weakly identifiable or approach a boundary, making direct inversion numerically unstable. L-BFGS-B addresses both issues by building a low-rank approximation $B_k^{-1} \approx \mathcal{H}_k^{-1}$ from the last m gradient difference pairs (s_j, r_j) and determining the search direction as $p_k = -B_k^{-1} \nabla \mathcal{L}(\theta_k)$.

The quasi-Newton approach is computationally efficient in strictly convex regimes, but it can fail in the presence of weak invertibility. To prove this, we must decompose the curvature matrix ($\mathcal{H}$) into its principal directions and magnitudes. We rewrite the Hessian as a sum of eigen-directions, each with an eigenvalue attached. The Hessian $\mathcal{H}$ at a point on a plateau admits an eigenpair $(\lambda_{ridge}, v_{ridge})$ with $|\lambda_{ridge}| \approx 0$, where v_{ridge} goes along the tangent direction of the simple curvature, causing the condition number to diverge:

$$\kappa(\mathcal{H}) = \frac{|\lambda_{max}|}{|\lambda_{min}|} \rightarrow \infty$$

The divergence breaks the algorithm's mechanics in two ways.

First, in the search direction update $p_k = -\mathcal{H}^{-1} \nabla \mathcal{L}$ (the complete analytical derivation of this LML gradient is provided in Appendix B.2), the inverse eigenvalue $1/\lambda_{min} \rightarrow \infty$ artificially amplifies infinitesimal numerical noise in the gradient along the ridge direction. The optimiser becomes hypersensitive to infinitesimal numerical perturbations in the direction of the plateau. To satisfy the Wolfe [16, 21] conditions and prevent the step size from exploding, the line search algorithm compensates by heavily truncating the step length α_k , even though the actual signal is basically flat. The algorithm prematurely truncates its step size and converges on a suboptimal local coordinate.

Second, the standard L-BFGS-B termination uses a stopping rule; the criterion triggers when the maximum component of the projected gradient falls below a predefined tolerance ϵ :

$$\max_i |(\text{proj} \nabla \mathcal{L}(\theta_k))_i| < \epsilon$$

On a flat ridge, the algorithm falsely registers convergence because the gradient is below that threshold. In reality, the surface is nearly flat along that direction. As the directional derivative vanishes ($\nabla \mathcal{L}(\theta) \cdot v_{ridge} \approx 0$), the optimiser mistakenly interprets this flat gradient as evidence of a true stationary point. It prematurely terminates the line search, declaring convergence at a suboptimal coordinate $\hat{\theta}_{local}$ rather than traversing the ridge to locate the global optimum. As will be empirically demonstrated in Chapter 4, this structural pathology causes point-estimated models to become highly sensitive to random initialisations, trapping them in local optima and yielding unpredictable, financially unstable volatility surfaces.

Chapter 4

The Pathology of Weak Identifiability

4.1 Mapping the LML Topology

To empirically assess why the quasi-Newton optimiser fails due to weak identifiability, we must conduct a holistic evaluation and map the exact objective landscape the algorithm navigates. For the target dates, we evaluate the Log-Marginal Likelihood (LML) function across a uniform mesh grid of the length-scale hyperparameters ℓ_K and ℓ_T . The signal and noise variances (σ_f^2, σ_n^2) were held constant near their optimal values.

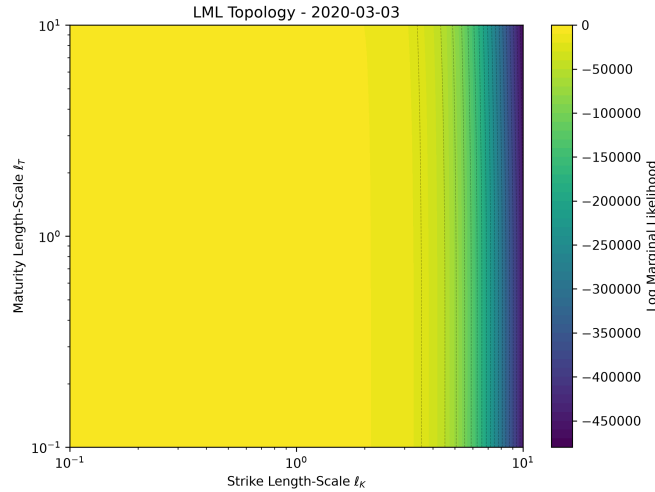


Figure 4.1: Topology on 3 March 2020.

For example, on 3 March 2020, the resulting contour in Figure 4.1 shows a pronounced lack of convexity. This pathology is exacerbated during high-stress market regimes, where the unique feature is the absence of a unimodal peak; the LML topography is dominated by an elongated plateau at the shorter length scale ℓ_K across the hyperparameter space.

A key consideration is the behaviour of the LML function at the lower bounds of the parameter space. A naive interpretation might suggest that the optimal choice of hyperparameters occurs at the strike length scale $\ell_K \rightarrow 0$. However, a near-zero length-scale implies that the correlation between two neighbouring points tends to zero, and

extrapolation vanishes entirely. This constitutes extreme overfitting, in which the model simply memorises microstructural noise, thereby losing its ability to generalise.

Because the landscape is mathematically flat along this ridge, the gradient vanishes ($\|\nabla\mathcal{L}\| \rightarrow 0$) and the exact Hessian matrix $\mathcal{H}$ becomes highly ill-conditioned (the formal quantification of this parameter uncertainty via the Fisher Information Matrix is derived in Appendix B.7). Specifically, the curvature parallel to the ridge is virtually zero, resulting in a corresponding eigenvalue $\lambda_{\text{ridge}} \approx 0$. As derived in Section 3.3, this causes the condition number to diverge, making the optimiser blind to the true global structure of the probability space.

4.2 Restart Instability

The ultimate choice of hyperparameters is heavily dependent on the initial position of the optimiser. The issue of the optimiser’s sensitivity to initialisation coordinates is made especially clear when navigating a flat ridge, as in Figure 4.1. The standard practice in Gaussian Process regression is to run the L-BFGS-B algorithm from multiple random starting points (restarts) and select the hyperparameter configuration that yields the highest LML [12, 18].

To understand why the initialisation coordinate carries such importance for the resulting estimate, we must mathematically break down the mechanics of the L-BFGS-B algorithm in the presence of a non-convex plateau. Let $\theta_0^{(1)}$ and $\theta_0^{(2)}$ be two distinct random initialisation vectors. The optimiser defines a descent direction p_k and relies on the strong Wolfe conditions to dictate the step length α_k :

$$\mathcal{L}(\theta_k + \alpha_k p_k) \leq \mathcal{L}(\theta_k) + c_1 \alpha_k \nabla \mathcal{L}(\theta_k)^\top p_k \quad (\text{Sufficient Decrease}) \quad (4.1)$$

$$|\nabla \mathcal{L}(\theta_k + \alpha_k p_k)^\top p_k| \leq c_2 |\nabla \mathcal{L}(\theta_k)^\top p_k| \quad (\text{Curvature Condition}) \quad (4.2)$$

where $0 < c_1 < c_2 < 1$. When the initialisation point $\theta_0^{(1)}$ lands on one side of the plateau, the algorithm rapidly descends the steep walls of the objective function, successfully satisfying the Wolfe conditions. However, the moment the trajectory reaches the flat ridge, the gradient magnitude collapses ($\|\nabla\mathcal{L}\| \rightarrow 0$) and the algorithm declares convergence.

At this point, the curvature condition (Equation 4.2) is the final stage in which the decision is made about whether to keep the step size α_k . When the algorithm descends into the valley, any step size yields a negligible change in the direction derivative $\nabla \mathcal{L}(\theta_k)$. In other words, neither the sufficient decrease nor the curvature conditions will be satisfied, leading the algorithm to converge falsely. Consequently, the final estimate $\hat{\theta}^{(1)}$ is entirely dependent on the specific location where the trajectory of $\theta_0^{(1)}$ first intercepted the ridge. Similarly, a different initialisation coordinate $\theta_0^{(2)}$ will intercept the plateau at a different location and terminate there. The resulting estimate of $\theta_0^{(n)}$ will yield different parameters;

none of them will be the optimal true maximum likelihood estimate, but rather just a random artefact of the optimiser's starting position.

Chapter 5

Bayesian Hyperparameter Inference via HMC

5.1 From Optimisation to Integration

In the previous chapter, we demonstrated that the topology of the log-marginal space for the financial volatility surfaces is quite ill-posed and features weak hyperparameter identifiability. This is because deterministic optimisation algorithms (e.g., L-BFGS-B) seek to maximise this objective function without accounting for the fact that the hyperparameter space cannot be collapsed to a single point estimate, which is not a definitive optimum. On the other hand, it is an arbitrary coordinate dictated by the optimiser’s initialisation and line-search termination criteria.

To address this structural fragility, this thesis examines the problem through the lens of integration rather than optimisation. Instead of attempting to find a single optimal estimate, we will adopt a fully Bayesian framework. This seeks to keep the full probability distribution over what those potential parameters could be, given the data. By applying Bayes’ theorem to the GP hyperparameters, we define the posterior distribution as:

$$p(\theta \mid \mathbf{y}, X) = \frac{p(\mathbf{y} \mid X, \theta)p(\theta)}{\int p(\mathbf{y} \mid X, \theta')p(\theta')d\theta'} \quad (5.1)$$

Where $p(\mathbf{y} \mid X, \theta)$ is the exact marginal likelihood defined in Chapter 3, and $p(\theta)$ is a joint prior distribution placed over the hyperparameters (such as the Matérn length-scales and noise variance) to constrain them to strictly positive, financially sensible bounds.

The advantage of this shift can be observed through forecasting. The standard point estimation methods rely on the rigid assumption that θ^* represents the absolute truth for predictions at unseen test inputs. Resulting in the condition predictive distribution $p(\mathbf{f}_* \mid X_*, X, \mathbf{y}, \theta^*)$. In contrast, the fully Bayesian predictive distribution marginalises out the hyperparameter uncertainty entirely by integrating the conditional predictions over the full posterior:

$$p(\mathbf{f}_* | X_*, X, \mathbf{y}) = \int p(\mathbf{f}_* | X_*, X, \mathbf{y}, \theta) p(\theta | \mathbf{y}, X) d\theta$$

In practical terms, this integral represents a continuous ensemble of Gaussian Processes. Rather than just picking one of them, which can lead to unstable extrapolation if the optimiser initialises in a suboptimal region of the parameter space. The Bayesian integral avoids this by hedging across all plausible explanations of the data. More specifically, the integral averages the predictive surfaces over all possible hyperparameter combinations, weighting them by their posterior probabilities. This approach handles cases where the data weakly identify whether a local market movement is driven by a short strike length scale (ℓ_K) or elevated signal variance (σ_f^2); the Bayesian predictive surface hedges between these competing scenarios. This structural averaging naturally avoids extreme localised extrapolations. Consequently, lowering the out-of-sample pricing error.

It is important to note that computing the exact Bayesian predictive distribution is practically impossible for a Gaussian Process regression. The primary reason for that is the denominator in 5.1, which lacks a closed-form analytical solution due to the highly nonlinear relationship between the hyperparameters and the inverted Matérn covariance matrix. As a consequence, this integral cannot be computed directly. Instead, we must approximate it using stochastic numerical methods. Hence, justifying the transition of analytical calculus to Markov Chain Monte Carlo (MCMC) simulation.

5.2 Markov Chain Monte Carlo

Having established that the computation of the Bayesian predictive integral exactly is analytically infeasible, we take the approach of Markov Chain Monte Carlo (MCMC) methods to construct an approximation of the posterior. The objective is to generate a range of dependent samples $\{\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(S)}\}$ such that, as $S \rightarrow \infty$, the empirical distribution converges to the exact target prior $p(\theta | \mathbf{y}, X)$.

The standard approach to this approximation problem is the Random Walk Metropolis (RWM) algorithm. Which takes the current parameter value and at each iteration step s , the algorithm suggests a nearby candidate θ' by adding isotropic Gaussian noise:

$$\theta' \sim \mathcal{N}(\theta^{(s)}, \Sigma_{prop})$$

where $\Sigma_{prop} = \sigma^2 I$ is the isotropic variance and controls the step size. The proposal is then accepted or rejected based on the Metropolis acceptance probability α :

$$\alpha = \min \left(1, \frac{p(\theta' | \mathbf{y}, X)}{p(\theta^{(s)} | \mathbf{y}, X)} \right)$$

The RWM algorithm is guaranteed to asymptotically converge to the true posterior

distribution in the limit of infinite iterations, but suffers severe operational degradation in high-dimensional spaces ($d \gg 1$).

Definition 5.1 (Typical Set). For a target posterior $p(\theta \mid y)$ over $\mathbb{R}^d$, the typical set $\mathcal{T}_\epsilon \subset \mathbb{R}^d$ is the narrow geometric shell that contains almost all probability mass, defined as the product of the likelihood density $p(\theta \mid y)$ and the expanding volume element $dV(r) \propto r^{d-1}dr$ [23]:

$$\mathcal{T}_\epsilon = \left\{ \theta \in \mathbb{R}^d : \left| -\frac{1}{d} \log p(\theta \mid y) - H(p) \right| < \epsilon \right\}. \quad (5.2)$$

As established in Definition 5.1, by the **concentration of measure** principle, the mode contains high probability density but virtually zero volume, whereas the outer parameter space contains vast volume but zero density. Consequently, almost all posterior mass concentrates in the thin hypersurface of the typical set $\mathcal{T}_\epsilon$. Because isotropic Gaussian proposals $q(\theta^* \mid \theta_k) = \mathcal{N}(\theta_k, \sigma^2 I)$ explore phase space randomly in all spatial directions, steps directed inward toward the mode or outward into vast volume fail to intersect this thin shell, yielding extremely low acceptance rates or near-zero exploration progress.

More specifically, the RWM relies on a gradient-free proposal distribution, in turn leading to a structural trade-off between the step size σ and the acceptance rate α . If the step size σ is chosen to be small, the candidate state remains within the typical set ($\alpha \approx 1$), and the exploration degrades into a diffusive random walk. After N steps, the distance scales like $\mathcal{O}(\sqrt{N})$, so the algorithm takes a long time to explore the full ridge geometry.

Conversely, if the steps taken by the algorithm are large. The isotropic candidate state will often be outside the narrow bounds of the typical set. Where the target density $p(\theta' \mid \mathbf{y}, X) \rightarrow 0$. Consequently, the acceptance probability α plummets, thereby paralysing the chain. Mathematically, for the acceptance probability ratio α to be nonzero, the size step must be scaled down by a factor of $\mathcal{O}(d^{-1/2})$ at the cost of stagnating the exploration of the surface in higher dimensions.

To avoid this diffusive, random-walk behaviour, we replace the isotropic, gradient-free candidate state model with a gradient-informed MCMC algorithm based on $\nabla_\theta \log p(\theta \mid \mathbf{y}, X)$. Which will actively steer the Markov chain along the contours of the typical set.

5.3 Hamiltonian Monte Carlo

A practical solution to the previous problem is Hamiltonian Monte Carlo (HMC), which reformulates the statistical sampling problem as a deterministic simulation of classical mechanics [19]. The algorithm is able to explore the high-dimensional typical set by introducing an auxiliary momentum variable and using information from the gradient of the target density to construct a vector field.

5.3.1 Phase Space and the Hamiltonian

Let the Gaussian Process hyperparameter vector θ map to a position vector $q \in \mathbb{R}^d$. To model this as a physical system, we introduce a fictitious, independent momentum vector $p \in \mathbb{R}^d$ representing an additional random push. The original parameter space is thus expanded into a $2d$ -dimensional **phase space** (q, p) . This approach allows us to treat the hyperparameters as a point moving in the mathematical space. Conceptually, the parameter space is augmented to simulate Hamiltonian dynamics, introducing an auxiliary momentum variable to enable vector-field dynamics without perturbing the target posterior over the original parameters. To achieve this, the joint distribution of the position and momentum variables is explicitly defined as statistically independent: $p(q, p) = p(q)p(p)$. Where the Hamiltonian $H(q, p)$ is the energy of this artificial system. In statistical mechanics, the canonical distribution relates the joint density to the energy function $H(q, p)$ [19]:

$$p(q, p) = \frac{1}{Z} \exp(-H(q, p))$$

where Z is the partition function. Hence, the joint distribution can be defined up to a constant. Because we defined the position and momentum to be independent, the Hamiltonian can be separated into Potential Energy $U(q)$ and Kinetic Energy $K(p)$:

$$H(q, p) = U(q) + K(p)$$

The potential energy $U(q)$ is defined as the negative logarithm of the unnormalized target posterior density under the Gaussian Process framework:

$$U(q) = -\log p(\mathbf{y} \mid X, q) - \log p(q)$$

The first term corresponds to the negative log-marginal likelihood of the Matérn 5/2 covariance structure that we defined in Chapter 3. The second term, $-\log p(q)$, accounts for the prior of the hyperparameters. Furthermore, the strictly positive Matérn parameters $\theta = \{\ell_K, \ell_T, \sigma_f, \sigma_n\}$ are mapped to the real line $q = \log(\theta)$ to ensure that the HMC particle traverses a smooth plane and doesn't diverge into numerical singularities during integration.

5.3.2 Hamiltonian Dynamics

The evolution of the system through phase space follows the rules of motion given by Hamilton's equations. They constitute a system of $2d$ coupled first-order ordinary differential equations that tell us how the state evolves over time.

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i} = [M^{-1}p]_i$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i} = -\frac{\partial U}{\partial q_i}$$

Mathematically, the first equation dictates that the position changes at a rate determined by the momentum. Hence, if momentum is larger in some direction, the parameter moves faster in that direction. While the other equation describes how the momentum changes according to the negative slope of the potential energy landscape. So, as the negative log-posterior increases, the particle drifts in some direction; the negative gradient pushes the particle back towards more probable regions, allowing it to rapidly traverse the flat ridges, mitigating the diffusive random-walk behavior characteristic of standard MCMC.

5.3.3 Theoretical Guarantees: Conservation, Volume, and Reversibility

In order for the Hamiltonian dynamics to be valid as a mechanism for a Markov chain, the continuous-time evolution must satisfy three strict conditions: energy conservation, volume preservation, and time reversibility. In other words, for the HMC to efficiently explore the surface's geometry, it must obey the three laws to ensure it does not perturb the target posterior distribution it samples from.

1. **Energy Conservation** The energy conservation rule keeps the Hamiltonian invariant, conserving energy over time.

Proof. By applying the multivariate chain rule and substituting Hamilton's equations, the time derivative becomes H :

$$\frac{dH}{dt} = \sum_{i=1}^d \left(\frac{\partial H}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial H}{\partial p_i} \frac{dp_i}{dt} \right) \quad (5.3)$$

$$\frac{dH}{dt} = \sum_{i=1}^d \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0$$

Because $\frac{dH}{dt} = 0$, the particle glides along isocontours of the joint density $p(q, p)$. Hence, the trajectory does not move toward higher or lower Hamiltonian values and remains at the same probability level rather than exploring them. In practice, we cannot integrate Hamilton's equations analytically, so we approximate the continuous dynamics using the Leapfrog method reviewed in the next sections. $\square$

2. **Volume Preservation (Liouville's Theorem)** Another fundamental condition is that Hamiltonian dynamics must preserve volume in phase space. Mathematically, this implies that a region of phase space mapped forward in time will retain its original volume.

Proof. Consider the vector field $\mathbf{V}$ governing the flow in phase space, defined as $\mathbf{V} = \left(\frac{dq}{dt}, \frac{dp}{dt}\right)$. The divergence of this vector field is:

$$\nabla \cdot \mathbf{V} = \sum_{i=1}^d \left(\frac{\partial}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial}{\partial p_i} \frac{dp_i}{dt} \right) \quad (5.4)$$

Substituting Hamilton's equations yields:

$$\begin{aligned} \nabla \cdot \mathbf{V} &= \sum_{i=1}^d \left(\frac{\partial}{\partial q_i} \left(\frac{\partial H}{\partial p_i} \right) + \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q_i} \right) \right) \\ &= \sum_{i=1}^d \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0 \end{aligned}$$

By Clairaut's theorem, the mixed second partial derivatives of H are equal, provided H is $\mathcal{C}^2$ continuous. Thereby, the two terms cancel, and the divergence vanishes ($\nabla \cdot \mathbf{V} = 0$). By Liouville's theorem [19], this phase-space volume preservation implies that the determinant of the Jacobian of the Hamiltonian flow is strictly equal to 1, completely bypassing the need to compute complex Jacobian determinants during the Metropolis acceptance step.

□

3. **Time Reversibility** The mapping is invariant under time reversal. If the system is simulated forward for time t , negating the momentum and simulating forward for time t returns the system to its exact initial state. This symmetry is mandatory for ensuring detailed balance in the resulting Markov chain.

Proof. Let the state of the system be defined by the coordinates (q, p) evolving under continuous time t . Consider the time-reversal transformation defined by mapping time $t \mapsto \tau = -t$ and simultaneously negating the auxiliary momentum $p \mapsto \tilde{p} = -p$, while position remains unchanged $q \mapsto \tilde{q} = q$.

Because the kinetic energy $K(p) = \frac{1}{2}p^\top M^{-1}p$ is a quadratic form, the Hamiltonian $H(q, p) = U(q) + K(p)$ is strictly symmetric with respect to momentum, ensuring that $H(q, p) = H(\tilde{q}, \tilde{p})$. Where, the kinetic energy $K(p) = \frac{1}{2}p^\top \mathbf{M}^{-1}p$ corresponds to the negative log-density of a Gaussian momentum prior, $p \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$, mirroring classical

kinetic energy $\frac{1}{2}mv^2$ with mass covariance matrix $\mathbf{M}$. Its quadratic symmetry ($K(p) = K(-p)$) guarantees the Hamiltonian invariance required for continuous-time reversibility.

Evaluating Hamilton's equations of motion under this transformation via the chain rule yields the position derivative:

$$\frac{d\tilde{q}}{d\tau} = \frac{dq}{d(-t)} = -\frac{dq}{dt} = -\frac{\partial H(q, p)}{\partial p}$$

Because the partial derivative of an even quadratic function is an odd function, evaluating $\frac{\partial H}{\partial p}$ at $-p$ yields $-\frac{\partial H}{\partial p}$. Substituting this into the previous equation results in:

$$\frac{d\tilde{q}}{d\tau} = -\left(-\frac{\partial H(\tilde{q}, \tilde{p})}{\partial \tilde{p}}\right) = \frac{\partial H(\tilde{q}, \tilde{p})}{\partial \tilde{p}}$$

Similarly, evaluating the evolution of the transformed momentum yields:

$$\frac{d\tilde{p}}{d\tau} = \frac{d(-p)}{d(-t)} = \frac{dp}{dt} = -\frac{\partial H(q, p)}{\partial q}$$

Because the potential energy $U(q)$ is independent of momentum, the spatial gradient remains completely unaffected by the sign of p :

$$\frac{d\tilde{p}}{d\tau} = -\frac{\partial H(\tilde{q}, \tilde{p})}{\partial \tilde{q}}$$

The transformed phase-space variables $(\tilde{q}, \tilde{p})$ perfectly satisfy the exact original Hamiltonian equations of motion with respect to reversed time τ . Therefore, the continuous-time Hamiltonian dynamics are strictly reversible, allowing the Metropolis-Hastings acceptance step to maintain detailed balance. $\square$

5.3.4 The No-U-Turn Sampler (NUTS)

While standard Hamiltonian Monte Carlo (HMC) eliminates the diffusive random-walk behavior of traditional Metropolis-Hastings algorithms, its practical efficiency depends critically on two user-defined parameters: the discrete integration step size ϵ and the trajectory length L (or total integration time $T = L\epsilon$).

In complex topological landscapes, such as the flat, weakly identifiable Log-Marginal Likelihood (LML) ridges of Gaussian Process volatility models, manually tuning L is problematic:

- **Suboptimal L (L too small):** The particle takes insufficient steps across phase space, causing the sampler to degrade into a high-variance, diffusive random walk.

- **Excessive L (L too large):** The particle follows a long Hamiltonian trajectory that eventually loops back onto itself (takes a "U-turn"), wasting computationally expensive $\mathcal{O}(N^3)$ Cholesky gradient evaluations without expanding the exploration of the posterior target density.

To resolve this tuning dilemma, Hoffman and Gelman [10] introduced the No-U-Turn Sampler (NUTS), an extension of HMC that dynamically determines the optimal trajectory length L at every iteration while strictly preserving time-reversibility and detailed balance.

Symplectic Integration and Dynamic Trajectories

At each iteration of the Markov chain, NUTS constructs a Hamiltonian trajectory starting from the current state $(\mathbf{q}, \mathbf{p})$ using the discrete-time, area-preserving *Leapfrog integrator*. Given a step size ϵ and mass matrix $\mathbf{M}$, a single leapfrog step updates position and auxiliary momentum as follows:

$$\mathbf{p}\left(t + \frac{\epsilon}{2}\right) = \mathbf{p}(t) - \frac{\epsilon}{2} \nabla_{\mathbf{q}} U(\mathbf{q}(t)), \quad (5.5)$$

$$\mathbf{q}(t + \epsilon) = \mathbf{q}(t) + \epsilon \mathbf{M}^{-1} \mathbf{p}\left(t + \frac{\epsilon}{2}\right), \quad (5.6)$$

$$\mathbf{p}(t + \epsilon) = \mathbf{p}\left(t + \frac{\epsilon}{2}\right) - \frac{\epsilon}{2} \nabla_{\mathbf{q}} U(\mathbf{q}(t + \epsilon)). \quad (5.7)$$

Rather than integrating forward for a fixed number of steps L , NUTS generates trajectories by building a recursive binary tree. At tree height $j \in \{0, 1, 2, \dots\}$, the algorithm selects a direction $v \in \{-1, +1\}$ uniformly at random and integrates 2^j leapfrog steps either backward ($v = -1$) or forward ($v = +1$) in time. This recursive doubling generates a trajectory spanning $2^{j+1} - 1$ candidate states, maintaining explicit reversibility.

The No-U-Turn Stopping Criterion

To determine when the trajectory begins to double back on itself, NUTS evaluates a geometric termination criterion. Let $\mathbf{q}_{\text{left}}$ and $\mathbf{q}_{\text{right}}$ denote the leftmost and rightmost position vectors of the candidate states within the current subtree, and let $\mathbf{p}_{\text{left}}$ and $\mathbf{p}_{\text{right}}$ be their respective momentum vectors.

The trajectory stops growing when the derivative of the squared Euclidean distance between $\mathbf{q}_{\text{left}}$ and $\mathbf{q}_{\text{right}}$ becomes negative. Mathematically, the expansion is halted as soon as:

$$\frac{d}{dt} \frac{1}{2} \|\mathbf{q}_{\text{right}} - \mathbf{q}_{\text{left}}\|^2 < 0. \quad (5.8)$$

By applying the chain rule, this condition is evaluated at both endpoints of the tree trajectory:

$$(\mathbf{q}_{\text{right}} - \mathbf{q}_{\text{left}})^\top \mathbf{p}_{\text{right}} < 0 \quad \text{or} \quad (\mathbf{q}_{\text{right}} - \mathbf{q}_{\text{left}})^\top \mathbf{p}_{\text{left}} < 0. \quad (5.9)$$

When the condition in Equation 5.9 is satisfied, continuing integration would pull the particle back toward its starting location. Tree expansion terminates immediately, and a new parameter state $\mathbf{q}^{(s+1)}$ is sampled from the candidate trajectory using a multinomial scheme weighted by the joint posterior density $\exp(-H(\mathbf{q}, \mathbf{p}))$, ensuring unbiased sampling from the exact target distribution.

Warmup Adaptation: Dual Averaging and Inverse Mass Matrix

To eliminate the need for manual tuning of the remaining hyperparameters, the integration step size ϵ and the diagonal mass matrix $\mathbf{M}$, NUTS utilizes a dual-phase adaptation protocol during the warmup stage (`num_warmup` = 500).

1. Step Size Adaptation via Dual Averaging: During warmup, NUTS adapts the integration step size ϵ using Nesterov’s dual-averaging scheme [10, 24]. The algorithm targets an optimal mean acceptance probability $\delta_{\text{target}} = 0.80$, an empirical trade-off benchmark established by Hoffman and Gelman [10] to maximize proposal distance while preventing rejections due to numerical integration.

$$H_m = \left(1 - \frac{1}{m + t_0}\right) H_{m-1} + \frac{1}{m + t_0} (\delta_{\text{target}} - \alpha_m), \quad (5.10)$$

$$\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} H_m, \quad (5.11)$$

where α_m is the observed acceptance probability of the m -th trajectory, $\mu = \log(10\epsilon_0)$, and γ, t_0 are stabilization parameters.

2. Mass Matrix Estimation: Simultaneously, the inverse mass matrix $\mathbf{M}^{-1}$ is continuously updated during the warmup phase by computing the sample covariance of the target parameters. In presence of anisotropic length-scales, estimating $\mathbf{M}^{-1}$ rescales the phase-space geometry, transforming elongated elliptical contours into isotropic hyperspheres.

Computational Implementation in JAX and NumPyro

In the implementation developed for this thesis, the NUTS engine is executed using the NumPyro library [17] backed by JAX [15]. This architecture affords significant mathematical and computational advantages when sampling over Gaussian Process hyperparameters:

- 1. Unconstrained Parameter Mapping and Weakly-Informative Priors:** To prevent numerical integration boundaries, the constrained positive kernel parameters

are mapped to unbounded real space $\mathbf{q} \in \mathbb{R}^4$:

$$\mathbf{q} = \begin{bmatrix} \log \ell_k \\ \log \ell_t \\ \log \sigma_f \\ \log \sigma_n \end{bmatrix} \in \mathbb{R}^4. \quad (5.12)$$

The latent GP model transforms these unbounded variables back via exponential maps ($\ell_k = \exp(q_1)$, etc.). To formulate the potential energy $U(\mathbf{q})$, weakly-informative Gaussian priors are evaluated in log-space: $q_1, q_2, q_3 \sim \mathcal{N}(0, 1)$ center the median prior expectation for length-scales and signal scale at $\exp(0) = 1.0$, while $q_4 \sim \mathcal{N}(-3, 1)$ embeds financial domain knowledge that market microstructure noise is small ($\exp(-3) \approx 0.05$).

2. **Automatic Differentiation through Cholesky Decompositions:** Evaluating the gradient $\nabla_{\mathbf{q}} U(\mathbf{q})$ at every leapfrog step requires differentiating through the $N \times N$ dense covariance matrix $\mathbf{K}_y = \mathbf{K}_{5/2} + \sigma_n^2 \mathbf{I}$. JAX compiles the anisotropic Matérn 5/2 kernel matrix computation and the subsequent Cholesky factorization ($\mathbf{L}\mathbf{L}^\top = \mathbf{K}_y$) directly into optimized XLA primitives, computing reverse-mode vector-Jacobian products (VJPs) with minimal overhead.
3. **Parallel Chain Execution:** By invoking `numpyro.set_host_device_count(2)`, two independent MCMC chains (`num_chains = 2`) are sampled in parallel using NumPyro [17]. Each chain undergoes 500 warmup iterations and 1000 sampling iterations, yielding 2000 posterior draws per date. Running multiple independent chains enables explicit MCMC convergence monitoring via the rank-normalized split- $\hat{R}$ metric [29]; a threshold of $\hat{R} < 1.05$ is strictly enforced across all hyperparameters, ensuring that between-chain variance remains under 5% of within-chain variance to confirm proper posterior mixing.

Chapter 6

Empirical Methodology and Experimental Design

6.1 Data Ingestion and Preprocessing

The empirical analysis utilises European-style option chains on the S&P 500 Index (SPX). Due to the sheer density of the SPX options market, evaluating the entire continuous surface simultaneously is computationally expensive and mathematically unnecessary for the sake of hyperparameter inference. Therefore, the data ingestion pipeline is designed to construct a representative discrete term structure.

The empirical analysis relies on institutional-grade market data sourced directly from the Chicago Board Options Exchange DataShop [36]. To ensure maximum precision and eliminate the microstructural unreliability common in free-tier data aggregators, historical end-of-day (EOD) SPX option chains were purchased and ingested in CSV format. A total of 21 dates were acquired, specifically stratified to test the HMC sampler across a spectrum ranging from high-distress volatility to sustained, low-volatility regimes.

A number of regimes were selected to test the HMC sampler, ranging from high-distress volatility to sustained-volatility regimes. The pipeline extracts a stratified cross-section of maturities, specifically targeting near-term (2 weeks), medium-term (5 and 12 weeks), and long-term (20 weeks) expirations. This provides the Gaussian Process with enough data to model a vast topological space that captures both steep, convex skew of short-dated options and the flatter, asymptotic behaviour of long-dated contracts.

For each maturity, the raw option chains are filtered using a strict preprocessing protocol to remove erroneous quotes and microstructure noise that will interfere with the construction of the Log-Marginal Likelihood landscape:

1. **Liquidity and Validity Constraints:** The market exhibits microstructure noise for various reasons, including, but not limited to, stale quotes and widening bid-ask spreads for OTM options. The noise changes the covariance matrix by adding a noise component (Equation 3.4). Thereby, altering conditioning, effective weighting,

and hyperparameter estimation. To prevent instability, we strictly filter the dataset. Any option with a trading volume of zero, or missing bid/ask quotes ($Bid \leq 0$ or $Ask \leq 0$), is discarded. Furthermore, quotes with extreme implied volatilities ($\sigma_{\text{imp}} \leq 0.01$ or $\sigma_{\text{imp}} \geq 2.5$) are removed, as these often are pricing errors or highly illiquid strikes.

2. **Coordinate Transformation:** A more natural coordinate system than the regular (K, T) is the $k = \log(K/S_0)$. The log-moneyness (see [II. Volatility & Surface Metrics](#)) represents a relative coordinate space that puts the strike price K into perspective with respect to the spot price S_0 . To ensure the Matérn kernel operates on a stable, scale-invariant domain, the strikes are mapped to log-moneyness $k = \log(K/S_0)$. Simultaneously, the expiration dates are converted into an annualised time-to-maturity metric T .
3. **Target Variable Construction:** As mentioned in Section [2.2](#), the target variable for the Gaussian Process is not the raw implied volatility, but rather the total implied variance. Following Gatheral’s SVI framework [\[8\]](#), the filtered implied volatilities are squared and scaled by maturity: $w(k, T) = \sigma_{\text{imp}}^2(k, T) T$. This ensures the calendar arbitrage monotonicity constraints detailed in Section [2.2.1](#) can be accurately evaluated.

6.2 Market Regimes

To ensure that the observed reduction in predictive error is attributable to the probabilistic methodology rather than to a particular feature of the market state, we evaluate both the deterministic point-estimate algorithm L-BFGS-B and the Bayesian HMC approaches across different environments. 21 dates spanning different market regimes were selected. Each regime was selected to stress-test a specific structural vulnerability in the surface fitting process:

- **High Volatility Shock (March 2020):** This regime is characterised by extreme market distress with the volatility index VIX exceeding 80 (i. e., the VIX index is a standardised metric representing the market’s expectation of 30-day forward-looking volatility). This month’s volatility surface exhibits severe backwardation (see [III. Arbitrage & Pricing Theory](#)) and microstructural friction, providing a rigorous test environment for the HMC sampler under difficult conditions. *Selected dates: 3, 13, 17, 19, 23, and 25 March 2020.*
- **Sustained Bear Market & Rate Shock (September – October 2022):** During this period, the volatility is stable and sustained. Additionally, the months of September and October feature a shifted yield curve (see [III. Arbitrage & Pricing](#)

[Theory](#)) and a steady loss in the equity market, making it a great fit to test the behaviour of the model under prolonged macroeconomic friction. *Selected dates: 16, 21, 26, 30 September and 5 of October 2022.*

- **Steady, Low-Vol Bull (Summer 2023):** The summer of 2023 serves as the baseline regime with low volatility and a standard volatility smile. This tests whether the GP model correctly relies on the parametric prior without overfitting the microstructural noise. *Selected dates: 7, 13, 21, 26, 30 of June 2023.*
- **High Momentum Expansion (January – February 2024):** Driven by aggressive, continuous market rallies. This regime specifically tests the pipeline’s ability to map right-tail asymmetry and the pronounced call skew (see [III. Arbitrage & Pricing Theory](#)) that emerges during rapid expansions. *Selected dates: 18, 24, 30 January and 9, 15 February 2024.*

6.3 Evaluation Metrics

To measure the performance of the competing inference methodologies, we must separate the evaluation framework into global predictive accuracy and local financial consistency. As established in [Section 2.2](#), a mathematically smooth interpolant is financially invalid if it permits arbitrage. Therefore, the models are evaluated across the following three numerical metrics:

1. **Out-of-Sample Predictive Accuracy (RMSE):** The Root Mean Square Error (RMSE) measures the out-of-sample accuracy to determine the global fit of the model, which is strictly evaluated on the testing set. Let N_{test} be the number of unseen option quotes, y_i be the observed total implied variance, and $\hat{y}_i$ be the model’s predictive mean. The out-of-sample RMSE is defined as:

$$\text{RMSE}_{\text{OOS}} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (y_i - \hat{y}_i)^2} \quad (6.1)$$

This metric penalises models that overfit the training data’s microstructural noise by measuring their generalisation error on unseen coordinates.

2. **Calendar Arbitrage Violations:** While [Section 2.2](#) defines the calendar arbitrage as a continuous monotonicity condition ($\partial w / \partial T \geq 0$), the numerical implementation is done using forward finite differences. Let $\mathbf{T} = \{T_1, T_2, \dots, T_m\}$ represent the sorted unique maturities. For each strike coordinate, the discrete derivative is computed and formally recorded as calendar arbitrage. These discrete derivatives

are computed and formally recorded as calendar arbitrage, summed across the grid whenever the discrete variance increment is negative:

$$\sum_{j=1}^{m-1} \mathbb{I} \left(\frac{w(k, T_{j+1}) - w(k, T_j)}{T_{j+1} - T_j} < 0 \right) \quad (6.2)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

3. **Butterfly Arbitrage Violations:** Similarly, the theoretical condition for the absence of butterfly arbitrage requires the risk-neutral probability density to remain non-negative, corresponding to a strictly convex call price curve ($\partial^2 C / \partial K^2 \geq 0$). To calculate it, we invert the predictive surface $w(k, T)$ to obtain European call prices $C(K, T)$ using the standard Black-Scholes pricing formula because the GP interpolates in total variance space. We evaluate the second derivative using a numerical approximation at the midpoint. Given a sorted grid of unique strike prices $\mathbf{K} = \{K_1, K_2, \dots, K_p\}$, let $\Delta C_i = C(K_{i+1}, T) - C(K_i, T)$ and $\Delta K_i = K_{i+1} - K_i$. The evaluation becomes:

$$\frac{\partial^2 C}{\partial K^2} \approx \frac{\frac{\Delta C_{i+1}}{\Delta K_{i+1}} - \frac{\Delta C_i}{\Delta K_i}}{\frac{1}{2}(\Delta K_{i+1} + \Delta K_i)} \quad (6.3)$$

The pipeline aggregates total butterfly violations by applying this finite-difference array operation and counting every instance in which the discrete second derivative falls below zero across all evaluated maturities.

Chapter 7

Empirical Results and Diagnostics

Previously, we established the theoretical foundations of deterministic point estimation in Chapter 4 and formalised the Bayesian Hamiltonian Monte Carlo methodology in Chapter 5. This chapter will present the empirical findings from the evaluation conducted across the varying market regimes defined in Section 6.2.

7.1 Predictive Accuracy: The Global Fit

The primary objective of any volatility surface interpolant is to accurately generalise to new market coordinates. The out-of-sample Root Mean Square Error (RMSE) is used to gauge the performance of this generalisation.

As established in Chapter 4.1 the deterministic optimiser exhibits weak identifiability on flat, non-convex ridges in the LML topological landscapes. The L-BFGS-B algorithm often stops prematurely at an arbitrary coordinate, subject to its initialisation point. Thereby, constraining the predictions of the GP and leads to severe overfitting of the model. Conversely, the Bayesian HMC framework integrates over the entire weakly identifiable ridge. By drawing samples from the full posterior distribution and computing a predictive mean as an expectation over the hyperparameter space, the HMC approach acts as a structural regulariser.

The observed results across different market regimes explicitly validate our hypothesis from Section 1.3, particularly during heavy market stress and turmoil. The largest reduction in error prediction was during the High Volatility Shock of March 2020. During this period, the deterministic optimiser was unable to generalise the surface correctly, resulting in a large OOS RMSE. As detailed in Figure C.1, on 25 March 2020, the Point-Estimated GP yielded a massive out-of-sample RMSE of 0.1038 (10.38%). Similarly, on 17 March 2020, the deterministic error spiked to 0.0802 (8.02%).

In stark contrast, the Bayesian HMC GP was able to contain the flat topological surface shown in Figure 7.1 during these exact same periods. By marginalising over the hyperparameter uncertainty, the HMC framework reduced the generalisation error on 25

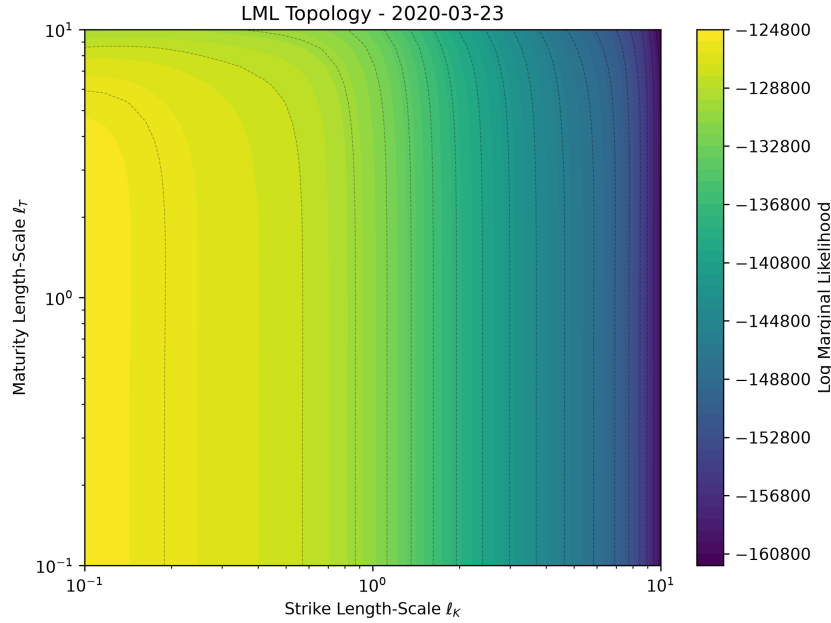


Figure 7.1: LML topology on 23 March 2023.

March 2020 to just 0.0263 (2.63%). A monumental $\sim 74\%$ reduction in predictive error compared to the deterministic baseline. On 23 March, the HMC error was at 0.0064, representing an $\sim 86\%$ improvement against the L-BFGS-B algorithm.

Date	Point GP RMSE	HMC GP RMSE	Error Reduction
2020-03-17	0.080205	0.016872	78.9%
2020-03-23	0.049056	0.006478	86.8%
2020-03-25	0.103873	0.026344	74.6%

Table 7.1: Out-of-Sample Predictive Accuracy during the March 2020 High Volatility Regime.

This severe reduction confirms that acknowledging and integrating over hyperparameter uncertainty actively prevents the kernel from collapsing into local optima. While the deterministic model aggressively fits the noisy, chaotic market data of a shifting regime, the probabilistic model maintains a mathematically sound global topology.

7.2 Visualising the Posterior Diagnostics

To understand how the Hamiltonian Monte Carlo sampler achieved the said error reductions, we must dig deeper and examine the empirical posterior distributions of the hyperparameters. As mentioned in Chapter 4, the LML space for financial volatility surfaces often features weakly identifiable, flat ridges. Instead of being perfectly convex, the optimum is a plateau.

Figure 7.2 illustrates the posterior distributions for the strike (ℓ_K) and maturity (ℓ_T) length-scales on 3 March 2020, a period of extreme market distress.

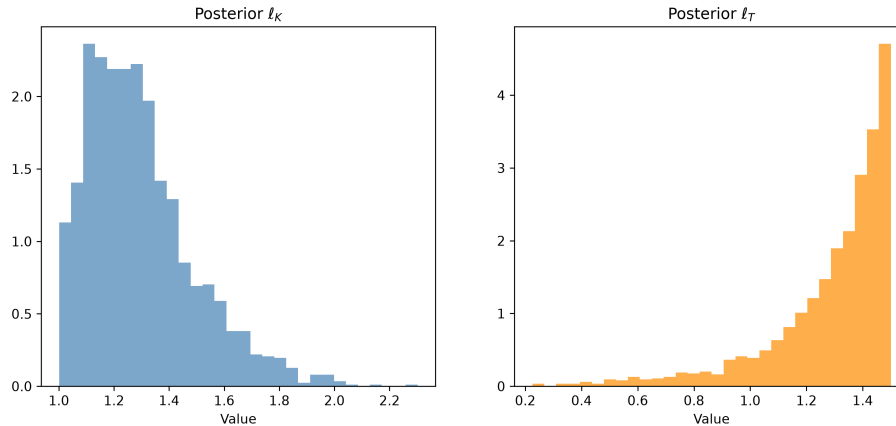


Figure 7.2: Posterior distributions of the Matérn length-scales during the High Volatility Shock (3 March 2020). The sampler achieves proper mixing, comprehensively exploring the weakly identifiable ridge rather than collapsing into localized boundary modes.

Unlike the deterministic optimiser, which prematurely terminates upon encountering flat gradients, the empirical posterior demonstrates that the HMC sampler successfully navigated and mapped the weakly identifiable space. As Figure 7.2 illustrates, the strike length-scale (ℓ_K) forms a continuous, unimodal distribution with a slight rightward skew, indicating that the sampler efficiently identified and mixed within a region of high probability mass. Simultaneously, the maturity length-scale (ℓ_T) exhibits a pronounced left-skew, systematically exploring a continuous range of plausible values while concentrating mass toward the upper end of its domain. This continuous exploration is a direct physical manifestation of the Hamiltonian dynamics interacting with the flat ridge topology mapped in Section 4.1. By utilizing auxiliary momentum, the simulated particle effectively traverses the phase space even when the gradient of the potential energy approaches zero ($\nabla U(q) \approx 0$). Consequently, the sampler maps the entire spectrum of hyperparameter uncertainty instead of stalling, providing a strong foundation for the structural averaging of the predictive surface.

While a deterministic optimiser prematurely terminates upon encountering these flat gradients, causing the restart instability documented in Section 4.2, the HMC algorithm successfully maps the entire ridge. Furthermore, the empirical data reveal that the location of this ridge is highly non-stationary across market regimes. During the 2020 COVID-19 shock, the market was very chaotic and exhibited high microstructural noise as the bid-ask spreads widened. The posterior distribution naturally favored a longer strike length-scale ($[1.1, 1.8]$), as can be seen from Figure 7.2, because a longer ℓ_K enforces strict structural regularisation. The Bayesian framework compensated for the degraded signal-to-noise ratio by enforcing strict structural regularisation, thereby maintaining an arbitrage-free surface.

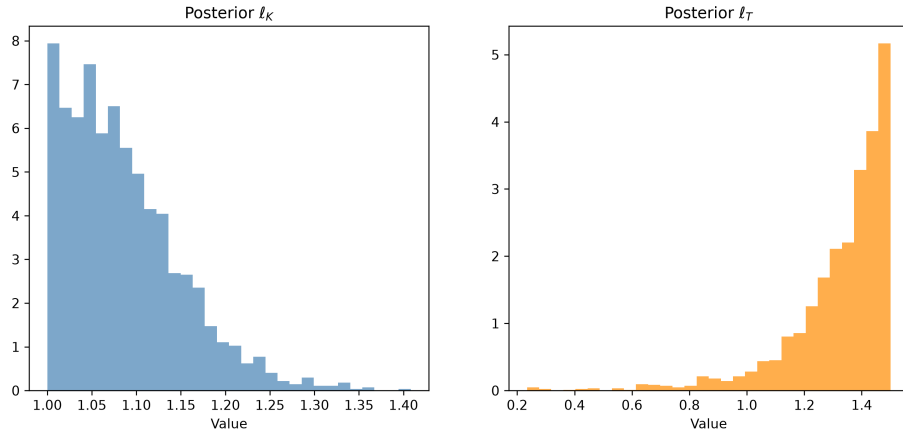


Figure 7.3: Posterior distributions of the Matérn length-scales during the High Momentum Expansion (15 February 2024). The sampler became more confident as the true economic shape was more defined.

In comparison, the early 2024 regime was in high-momentum expansion. During aggressive, continuous market rallies, option markets develop very sharp, pronounced structural skews (particularly on the call side (see [III. Arbitrage & Pricing Theory](#)) as traders chase the rally). Rather than the HMC shifting, the peak of the marginal posterior shifted to a lower length-scale ($[1.0, 1.15]$), as can be seen in [Figure 7.3](#), because the data had a cleaner, stronger signal, and the true economic shape of the surface was much steeper. A shorter length scale allows the model to be more flexible and detailed.

By using structural averaging, the framework mitigates the influence of parameter extremal behaviours of specific length scales. In other words, it effectively hedges between the mathematical extremes of a weakly identifiable region. This prevents the kernel from memorising microstructural noise or making the surface too rigid.

7.3 Financial Consistency and the Arbitrage Trade-off

While the out-of-sample RMSE measures how well the model extrapolates across the macroscopic surface, it does not guarantee economic validity. As mentioned in [Section 2.2](#), a mathematically smooth interpolant is financially inadmissible if it permits arbitrage. Both frameworks were evaluated for financial consistency with respect to the calendar and butterfly arbitrage constraints.

The empirical results, aggregated in [Appendix C.1](#), reveal the mostly consistent structural flaw in the deterministic point-estimation framework across regimes. By aggressively maximising the log-marginal likelihood, the L-BFGS-B optimiser frequently overfits the microstructural noise of the bid-ask spreads. This is especially evident during the 2024 regime, mentioned in [Section 7.1](#). Additionally, [Figure 7.4](#) provides insight into

how both algorithms fit the surface. It is evident that although the economic structure is similar, the way each algorithm represents it is drastically different.

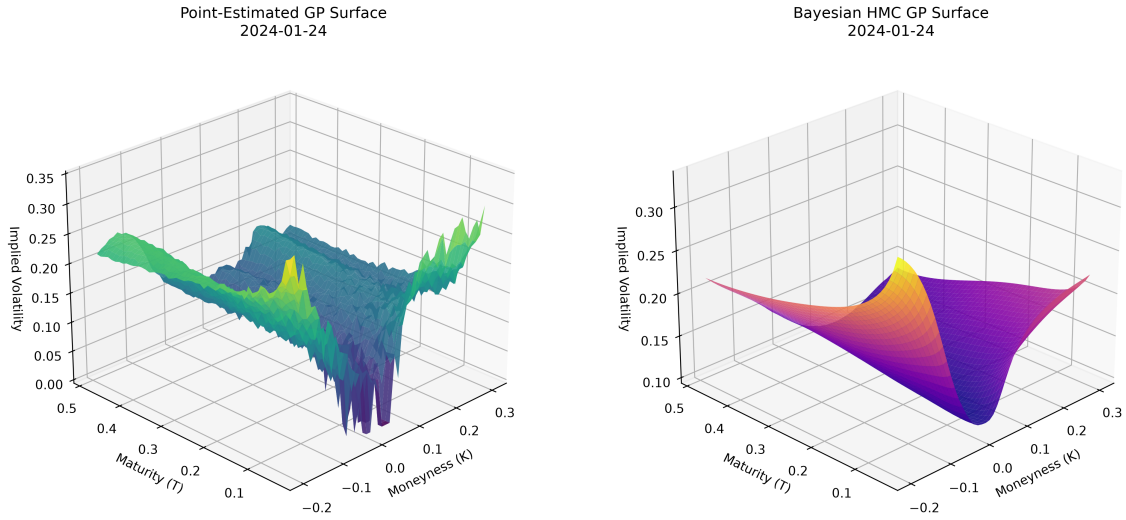


Figure 7.4: 3D implied volatility surface comparison on 2024-01-24. The deterministic L-BFGS-B model overfits local noise into arbitrage-violating non-convexities, whereas full Bayesian HMC marginalisation guarantees a smooth, structurally consistent predictive surface.

Consequently, overfitting the microstructural noise induces high-frequency non-convexities across the predictive surface. This is especially pronounced in Figure 7.4. When inverted to call prices, these strict convexities violate the $\partial^2 C / \partial K^2 \geq 0$ condition required by the Breeden-Litzenberger theorem, resulting in negative risk-neutral probabilities.

An analysis of the 21 unique historical trading days reveals that the Point-Estimated GP generated butterfly arbitrage violations on **61.9%** of the evaluated dates. More critically, the model regularly suffered from simultaneous, systemic collapses of both static arbitrage conditions:

- Across the four market regimes, the deterministic point estimator simultaneously violated *both* calendar and butterfly constraints on 14.3% of the evaluated days.
- During periods of extreme volatility (March 2020), this joint-failure rate escalated, with simultaneous violations spiking as high as 223 calendar and 127 butterfly errors on 19 March 2020.
- Even in the low-volatility Bull market (Summer 2023), the deterministic model suffered catastrophic failures, including a peak of 450 calendar violations and 113 butterfly violations on 21 June 2023.

Table 7.2 illustrates that during rapid regime shifts and volatile environments, the point estimator shortens the length scales and memorises the entire surface, nominally

Metric	Point-Estimated GP	Bayesian HMC GP
Clean Calendar Days	85.7% (18/21)	71.4% (15/21)
Clean Butterfly Days	38.1% (8/21)	61.9% (13/21)
Dual-Violation Days	14.3% (3/21)	9.5% (2/21)

Table 7.2: Aggregate Arbitrage Failure Rates across the 21 unique historical trading days.

beating the HMC GP on certain days, but trading financial validity for accuracy. The mechanics of this dangerous accuracy-consistency trade-off will be explicitly examined in the following section.

The Bayesian HMC framework changes the perspective on the problem entirely, instead of finding the “best” single hyperparameter. The NUTS algorithm generates samples across the full posterior distribution, successfully exploring the entire weakly identifiable ridge surrounding the LML optimum. By computing the predictive expectation over this prior-constrained hyperparameter space, the Bayesian framework itself acts as an automated structural regulariser. As the algorithm averages the predictive surfaces generated by varying length-scales, the random, chaotic microstructural noise cancels itself out. The resultant marginalised expectation isolates the macroeconomic volatility skew and unveils its true macroeconomic landscape.

To summarise, the Bayesian HMC framework demonstrated strong financial validity and overall robustness with respect to structural regularisation. During the severe out-of-distribution coordinate expansion of June 2023, while the model could not completely overcome the extreme structural contradictions present in the extreme out-of-distribution observations, it drastically reduced absolute violations (e.g., dropping calendar violations from 450 down to 37, and completely eradicating butterfly violations from 113 down to 0 on 21 June).

7.4 The Accuracy-Consistency Trade-off

During highly liquid, low-volatility market environments, the deterministic point estimator frequently achieves a nominally superior out-of-sample RMSE compared to the Bayesian ensemble. However, this performance can be deceiving if we do not consider the overall picture.

7.4.1 The Illusion of Deterministic Superiority

To understand the mechanics of this illusion, we must examine the behaviour of the quasi-Newton optimiser on a highly identifiable Log-Marginal Likelihood (LML) landscape. During steady market regimes, the volatility surface is generally smooth and structurally stable. Because the broader shape of the surface is simple, the residual data is largely

dominated by microstructural market friction and not necessarily by economic patterns. In which case, the held-out points may contain very similar quote noise, so a model that partly memorises that noise can still score a very low RMSE but be economically incorrect.

7.4.2 Empirical Case Studies: The 2022 and 2023 Regimes

To explicitly outline this behaviour of the deterministic point estimator, we will consider the SPX options market on **5 October 2022** and **13 June 2023** for empirical evaluation. On both dates, the out-of-sample RMSEs were lower than their Bayesian counterparts, a direct result of the point estimator successfully locking onto the exact coordinates of the noisy training data.

Date	Method	OOS RMSE (IV)	C. Viol.	B. Viol.
2022-10-05	Point-Estimated GP	0.002564	0	101
	Bayesian HMC GP	0.003105	0	10
2023-06-13	Point-Estimated GP	0.022153	0	77
	Bayesian HMC GP	0.024853	0	0

Table 7.3: The Accuracy-Consistency Trade-off, where C. Viol. and B. Viol. represent the calendar and butterfly arbitrage violations respectively. The Point-Estimated GP achieves a slightly lower RMSE by overfitting microstructural noise, generating dozens of arbitrage violations.

As detailed in Table 7.3, on 13 June 2023, the Point-Estimated GP achieved an RMSE of just 0.022153. However, it achieved this low score at the expense of the surface's financial stability, resulting in **77 butterfly arbitrage violations**. Similarly, on 5 October 2022, the deterministic model outperformed the Bayesian approach by 0.00054 in RMSE. But it had to compromise on financial arbitrage, generating **101 butterfly violations**.

This smoothing effect explains the HMC results on 13 June 2023. By refusing to overfit the local microstructural noise, the Bayesian surface slightly "detuned" itself from the exact coordinates of the test set, resulting in a microscopic increase in generalisation error (RMSE of 0.024853). However, this marginal RMSE penalty (≈ 0.0027) was the mathematical price required to eliminate the localised structural anomalies, dropping the butterfly violations from 77 down to exactly 0.

The empirical evidence therefore confirms the central hypothesis of this dissertation. Evaluating non-parametric volatility models only on OOS RMSE creates a financial illusion of superiority. By replacing point estimation with Bayesian hyperparameter inference, we impose a mandatory regularisation mechanism that explicitly prioritises risk-neutral probability constraints over the memorisation of market friction.

Chapter 8

Conclusion

8.1 The Pathology of Point Estimation

This dissertation set out to resolve the structural instability present in non-parametric volatility surface models, specifically addressing the failure of the deterministic point estimation approach to Gaussian Processes during market turmoil. The core mathematical discovery of this research is that Log-Marginal Likelihood (LML) for options is often non-convex, featuring a weakly identifiable hyperparameter space. This directly undermines the standard assumption that a single, optimal set of hyperparameters can be isolated from the market data.

Most importantly, this topological mapping reveals a structural flaw, not in the deterministic L-BFGS-B optimiser, but within the Maximum Likelihood Estimation (MLE) objective it is instructed to solve. The L-BFGS-B algorithm is a highly efficient quasi-Newton method; it mathematically succeeds in finding the exact coordinates that maximise the probability of the observed training data. However, in the context of chaotic financial markets, the pursuit of an absolute optimum yields a spurious objective. The absolute optimum corresponds to shorter length scales, where the LML score is highest because it aggressively memorises in-sample microstructural friction, such as bid-ask spreads and asynchronous tick sizes. Consequently, this achieves a nominally lower OOS RMSE in certain regimes.

Therefore, by restricting the model to collapse its hyperparameter uncertainty into a single noise-fitting coordinate on a flat ridge, point estimation transforms the Gaussian Process from generalisable, out-of-sample forecasting into a highly brittle, overfitted function. Directly contradicting the fundamental goal of MLE.

8.2 The Accuracy-Consistency Trade-off

The consequence of this overfitting introduces a dangerous paradigm into quantitative trading: the financial illusion of predictive accuracy. A low generalisation error becomes

meaningless unless accompanied by strict arbitrage constraints.

The empirical findings of this research highlight the instability of the deterministic point estimation approach under these economic constraints. Across the evaluated historical data, the point estimator achieved a 61.9% butterfly arbitrage failure rate. Additionally, this instability seldomly escalated into systemic dual-violations, failing both calendar and butterfly conditions on 14.3% of the evaluated dates, with peak failures occurring during the severe market shocks of March 2020 and the summer of 2023. Thereby supporting the claim that the pursuit of the absolute maximum likelihood frequently guaranteed negative risk-neutral probabilities.

Conversely, the Bayesian framework adopts the role of a structural regulariser; meanwhile, the No-U-Turn Sampler (NUTS) effectively samples across the entire weakly identifiable posterior ridge to approximate this high-dimensional integral, generating an ensemble of predictive surfaces. By averaging them, the fatal structural flaw of the point estimator is resolved. Furthermore, by replacing point estimation with Hamiltonian Monte Carlo (HMC) inference, the chaotic microstructural noise cancels itself out due to the marginalisation process of the framework.

Ultimately, this regularisation exposes an accuracy-consistency trade-off between out-of-sample error and financial validity. This compromise is especially noticeable during steady or low-volatility regimes such as October 2022 and June 2023. The Bayesian model accepts a small penalty in generalisation error compared to the deterministic optimiser. However, this marginal reduction in accuracy is the mandatory price for an economically admissible surface, which guarantees the eradication of structural arbitrage.

8.3 Computational Limitations and Future Work

While the Bayesian Hamiltonian Monte Carlo framework decisively outperforms the deterministic point estimation approach in preserving economic admissibility, its deployment and upkeep during live trading sessions are heavily constrained by computational complexity. Although the time complexity of both algorithms is $\mathcal{O}(N^3)$, the fundamental mathematical bottleneck is the computationally expensive Cholesky decomposition of the covariance matrix. The paramount difference between them comes from the constant factor, which represents the number of times the algorithms have to execute this expensive $\mathcal{O}(N^3)$ decomposition. Based on the hyperparameter configurations used in this study, evaluating the full posterior via Hamiltonian Monte Carlo requires roughly 1,000 times more computational overhead than deterministic point estimation.

Additionally, due to the time constraints of this research, thorough MCMC diagnostics, such as trace plots to assess mixing, autocorrelation functions (ACF), and Effective Sample Size (ESS), are omitted from this text. However, preliminary $\hat{R}$ metrics indicated stable convergence.

It is reasonable to assert that the current HMC implementation is most optimally suited for end-of-day settlement pricing (i. e., the official daily valuation of derivative contracts), risk management, and portfolio margin calculations (i.e., the determination of regulatory capital requirements based on aggregate portfolio risk). In this scenario, the extreme accuracy and strictly arbitrage-free surfaces are important, but execution latency is a secondary concern. Furthermore, the current computational complexity of the NUTS sampler [10] is expensive and impractical for High Frequency Trading (HFT), where volatility surfaces must be updated on sub-millisecond timescales.

The logical progression for future quantitative research is to bridge this computational gap by exploring Variational Inference (VI) or Sparse Gaussian Processes. By framing Bayesian inference as an optimisation problem rather than a stochastic sampling problem, VI has the potential to approximate the true Bayesian posterior in a fraction of the time. Successfully implementing a Variational Gaussian Process could theoretically maintain the powerful regularising benefits of hyperparameter ensembling while achieving the near-deterministic execution speeds required for high-frequency trading floors.

Furthermore, amortised inference should be considered a logical progression from variational approximations. To bypass the computational bottleneck of the Cholesky decomposition, the intensive Hamiltonian Monte Carlo pipeline can be executed entirely offline. In this framework, a deep neural network is trained to map live microstructural market inputs, such as asynchronous bid-ask spreads (i.e., bid and ask prices updated at different timestamps), directly to the regularised posterior predictive surface. This fundamental shift would effectively reduce the online time complexity from $\mathcal{O}(N^3)$ down to a strictly $\mathcal{O}(1)$ deterministic forward pass. For quantitative infrastructures that require the exact analytical Gaussian Process rather than a neural surrogate, future implementations should explore direct hardware-level acceleration. By implementing the exact Cholesky factorisation directly on high-performance GPUs using C++ CUDA kernels, the operational overhead of Python abstractions could be directly circumvented. While preserving mathematical precision while drastically minimising execution latency.

Appendix A

Pipeline Architecture

To ensure full computational reproducibility, the codebase developed for this dissertation, including the Hamiltonian Monte Carlo implementation and the Arbitrage Auditor, is publicly available at: [Masters Dissertation Project](#).

- `01_ingest_spx.py`: data ingestion,
- `02_preprocess.py`: filtering, total variance construction, train/test split,
- `03_prior_model.py`: parametric prior,
- `04_gp_point_estimation.py`: point-estimated GP,
- `05_topology_scan.py`: LML topology analysis,
- `06_restart_sensitivity.py`: restart instability diagnostics,
- `07_gp_hmc.py`: Bayesian inference with HMC/NUTS [10, 19] executed via NumPyro [17] and JAX [15],
- `08_compare_results.py`: comparative evaluation,
- `09_plots.py`: figures and posterior visualisation.

Appendix B

Mathematical Proofs and Extended Derivations

In this Appendix, we provide the mathematical derivations and continuous-time calculus required to make the algorithmic framework discussed in the main text rigorous. While the foundational theorems are established in standard mathematical and financial literature, the step-by-step expansions, matrix calculus, and specific applications to the Matérn kernel presented here have been manually derived and adapted for the exact structural constraints of this dissertation.

For full transparency and to demonstrate the independent execution of these mathematical mechanics, the original handwritten derivations corresponding to this appendix have been digitized and made available in the project’s repository at [Masters Dissertation Project](#).

B.1 The Schur Complement and the GP Predictive Distribution

This derivation provides the formal proof of how a Gaussian Process conditions its predictions on observed market data using a standard block-matrix inversion approach outlined by Rasmussen and Williams [12]. By applying the Schur complement to a partitioned block covariance matrix, we isolate the exact conditional mean and covariance for unseen test coordinates.

B.1.1 The Setup: The Joint Multivariate Gaussian

Proof. We begin by defining the joint probability distribution of our observed, noisy market targets (Total Implied Variance, denoted as y) and our unobserved, noise-free predictive targets denoted as f_* . The joint distribution becomes multivariate Gaussian, assuming the data has been scaled or the parametric prior has been subtracted:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} K(X, X) + \alpha I & K(X, X_*) \\ K(X_*, X) & K(X_*, X_*) \end{bmatrix} \right)$$

For algebraic clarity, we will denote the covariance matrix into blocks as follows:

- $K_{yy} = K(X, X) + \alpha I$ (The $n \times n$ training covariance with the regularization nugget α)
- $K_{y*} = K(X, X_*)$ (The $n \times n_*$ cross-covariance between training and test inputs)
- $K_{*y} = K(X_*, X)$ (The $n_* \times n$ cross-covariance, which is K_{y*}^T)
- $K = K(X_*, X_*)$ (The $n_* \times n_*$ prior test covariance)

Thus, the joint covariance matrix is $\Sigma = \begin{bmatrix} K_{yy} & K_{y*} \\ K_{*y} & K \end{bmatrix}$.

B.1.2 The Derivation: Affine Transformation and the Schur Complement

To find the conditional distribution $P(f_* | y)$, we must mathematically isolate f_* from y . We achieve this by applying a linear affine transformation to decorrelate the variables. Firstly, we define a transformation matrix M designed to eliminate the cross-covariance terms:

$$M = \begin{bmatrix} I & \mathbf{0} \\ -K_{*y}K_{yy}^{-1} & I \end{bmatrix}, Z = \begin{bmatrix} y \\ f_* \end{bmatrix}$$

We apply this transformation matrix to our joint random vector Z to obtain:

$$MZ = M \begin{bmatrix} y \\ f_* \end{bmatrix} = \begin{bmatrix} y \\ f_* - K_{*y}K_{yy}^{-1}y \end{bmatrix}$$

To calculate the new covariance after a linear transformation of a random vector Z which follows a multivariate Gaussian distribution with Covariance Σ . We compute the covariance of this new transformed system using the property $\text{Cov}(MZ) = M\Sigma M^T$:

$$M\Sigma M^T = \begin{bmatrix} I & \mathbf{0} \\ -K_{*y}K_{yy}^{-1} & I \end{bmatrix} \begin{bmatrix} K_{yy} & K_{y*} \\ K_{*y} & K \end{bmatrix} \begin{bmatrix} I & -K_{yy}^{-1}K_{y*} \\ \mathbf{0} & I \end{bmatrix} \quad (\text{B.1})$$

Expanding the matrix multiplication step-by-step, we multiply M by Σ :

$$M\Sigma = \begin{bmatrix} K_{yy} & K_{y*} \\ -K_{*y}K_{yy}^{-1}K_{yy} + K_{*y} & -K_{*y}K_{yy}^{-1}K_{y*} + K \end{bmatrix} = \begin{bmatrix} K_{yy} & K_{y*} \\ \mathbf{0} & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix}$$

Next, we multiply by M^T :

$$\begin{aligned}
 M\Sigma M^T &= \begin{bmatrix} K_{yy} & K_{y*} \\ \mathbf{0} & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} \begin{bmatrix} I & -K_{yy}^{-1}K_{y*} \\ \mathbf{0} & I \end{bmatrix} \\
 M\Sigma M^T &= \begin{bmatrix} K_{yy} & -K_{yy}K_{yy}^{-1}K_{y*} + K_{y*} \\ \mathbf{0} & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} = \begin{bmatrix} K_{yy} & \mathbf{0} \\ \mathbf{0} & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} \quad (\text{B.2})
 \end{aligned}$$

B.1.3 Decorrelation and Independence

The resulting $M\Sigma M^T$ block from the B.2 linear transformation gives exactly $\mathbf{0}$ in the off-diagonal covariance blocks. Since y and $(f_* - K_{*y}K_{yy}^{-1}y)$ are jointly Gaussian, they become strictly statistically independent and stripped of any correlation. We can formally define the term in the bottom right block as the **Schur Complement** of K_{yy} in the joint matrix Σ :

$$S = K_{**} - K_{*y}K_{yy}^{-1}K_{y*}$$

Because $f_* - K_{*y}K_{yy}^{-1}y$ is independent of y , its conditional distribution given y is identical to its marginal distribution:

$$f_* - K_{*y}K_{yy}^{-1}y \mid y \sim \mathcal{N}(\mathbf{0}, S)$$

B.1.4 The Result: The Conditional Predictive Distribution

To find the distribution of our target predictive variable f_* conditioned on the observed market data y , we simply rearrange the isolated term. Since y is treated as a known constant when conditioning:

$$f_* \mid y \sim \mathcal{N}(K_{*y}K_{yy}^{-1}y, S)$$

By substituting our original notation back into the equation, we yield the explicit, exact formulas for the Gaussian Process predictive mean vector ($\bar{f}_*$) and predictive covariance matrix (Σ_*):

Predictive Mean:

$$\bar{f}_* = K(X_*, X)[K(X, X) + \alpha I]^{-1}y \quad (\text{B.3})$$

Predictive Covariance:

$$\Sigma_* = K(X_*, X_*) - K(X_*, X)[K(X, X) + \alpha I]^{-1}K(X, X_*) \quad (\text{B.4})$$

□

B.2 Matrix Calculus of the Log-Marginal Likelihood Gradients

While the generic formulation of the log-marginal likelihood gradients is well-established in spatial statistics [12], this derivation provides the formal proof of how gradient-based optimisers navigate the landscape and why they stall on flat ridges by computing the analytical partial derivatives of the Log-Marginal Likelihood (LML) with respect to GP covariance kernel hyperparameters θ_j (such as length-scales ℓ_K, ℓ_T , signal variance σ_f^2 , or noise variance σ_n^2).

B.2.1 The Setup: The Objective Function

Proof. Let $y \in \mathbb{R}^n$ denote the vector of observed targets (Total Implied Variance) and $X \in \mathbb{R}^{n \times d}$ denote the input matrix. Hence, $p(y | X, \theta)$ is the probability of seeing the financial targets y , given our input coordinates X and chosen hyperparameters θ . Following the standard convention, we assume a zero-mean Gaussian Process prior (or residuals after subtracting a parametric mean prior) [12]. Thereby, the standard Multivariate Gaussian PDF is:

$$p(y | X, \theta) = (2\pi)^{-\frac{n}{2}} |K_y|^{-\frac{1}{2}} \exp\left(-\frac{1}{2} y^T K_y^{-1} y\right)$$

where $K_y = K_\theta + \sigma_n^2 I \in \mathbb{R}^{n \times n}$ represents the noisy covariance (Gram) matrix. By taking the natural logarithm, we obtain the Log-Marginal Likelihood objective function $\mathcal{L}(\theta) = \log p(y | X, \theta)$ that we explicitly defined in Section 3.3.

$$\mathcal{L}(\theta) = \underbrace{-\frac{1}{2} y^T K_y^{-1} y}_{\text{Data Fit Term}} - \underbrace{\frac{1}{2} \log |K_y|}_{\text{Complexity Penalty}} - \underbrace{\frac{n}{2} \log(2\pi)}_{\text{Normalization Constant}} \quad (\text{B.5})$$

B.2.2 Mathematical Prerequisites: Matrix Calculus Rules

First, we invoke two fundamental matrix calculus identities in order to differentiate $\mathcal{L}(\theta)$ with respect to a scalar hyperparameter θ_j :

Identity 1: Derivative of an Inverse Matrix

Using the standard product rule, we differentiate the invertible matrix K_y with respect to a scalar θ_j :

$$\frac{\partial}{\partial \theta_j} (K_y K_y^{-1}) = \frac{\partial K_y}{\partial \theta_j} K_y^{-1} + K_y \frac{\partial K_y^{-1}}{\partial \theta_j} = \mathbf{0} \quad (\text{B.6})$$

From Equation B.6, we multiply the left by K_y^{-1} and isolate $\frac{\partial K_y^{-1}}{\partial \theta_j}$ to obtain:

$$\frac{\partial K_y^{-1}}{\partial \theta_j} = -K_y^{-1} \frac{\partial K_y}{\partial \theta_j} K_y^{-1}$$

Identity 2: Jacobi's Formula for the Log-Determinant

Jacobi's formula states that for any differentiable matrix K_y , its derivative $\frac{\partial}{\partial \theta_j} |K_y|$ is equal to $|K_y| \text{Tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_j} \right)$. Specifically, when we apply the chain rule to the logarithm of the determinant, we obtain the steps B.7 through simplification.

$$\frac{\partial}{\partial \theta_j} \log |K_y| = \frac{1}{|K_y|} \frac{\partial |K_y|}{\partial \theta_j} = \frac{1}{|K_y|} \left[|K_y| \text{Tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_j} \right) \right] = \text{Tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_j} \right) \quad (\text{B.7})$$

B.2.3 The Derivation: Differentiating $\mathcal{L}(\theta)$

We take the partial derivative of $\mathcal{L}(\theta)$ with respect to θ_j component-by-component:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \left(-\frac{1}{2} y^T K_y^{-1} y \right) - \frac{\partial}{\partial \theta_j} \left(\frac{1}{2} \log |K_y| \right) - \frac{\partial}{\partial \theta_j} \left(\frac{n}{2} \log(2\pi) \right)$$

Step A: Differentiating the Data Fit Term

Since y does not depend on θ_j , we pass the partial derivative operator inside:

$$\frac{\partial}{\partial \theta_j} \left(-\frac{1}{2} y^T K_y^{-1} y \right) = -\frac{1}{2} y^T \left(\frac{\partial K_y^{-1}}{\partial \theta_j} \right) y$$

Substituting Identity B.6 into the equation:

$$-\frac{1}{2} y^T \left(-K_y^{-1} \frac{\partial K_y}{\partial \theta_j} K_y^{-1} \right) y = \frac{1}{2} y^T K_y^{-1} \frac{\partial K_y}{\partial \theta_j} K_y^{-1} y$$

To simplify the term into a quadratic form, we define the predictive weights vector $\alpha = K_y^{-1} y \in \mathbb{R}^n$:

$$\frac{\partial}{\partial \theta_j} \left(-\frac{1}{2} y^T K_y^{-1} y \right) = \frac{1}{2} \alpha^T \frac{\partial K_y}{\partial \theta_j} \alpha$$

Step B: Differentiating the Complexity Penalty Term

Using Identity 2 directly:

$$\frac{\partial}{\partial \theta_j} \left(\frac{1}{2} \log |K_y| \right) = \frac{1}{2} \text{Tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_j} \right)$$

Step C: Differentiating the Constant

Since $\frac{n}{2} \log(2\pi)$ is independent of θ_j , this term is equal to 0:

$$\frac{\partial}{\partial \theta_j} \left(\frac{n}{2} \log(2\pi) \right) = 0$$

B.2.4 The Result: The Gradient Equation

Combining Step A, Step B, and Step C, we obtain the exact gradient of the Log-Marginal Likelihood with respect to hyperparameter θ_j :

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \frac{1}{2} y^T K_y^{-1} \frac{\partial K_y}{\partial \theta_j} K_y^{-1} y - \frac{1}{2} \text{Tr} \left(K_y^{-1} \frac{\partial K_y}{\partial \theta_j} \right) \quad (\text{B.8})$$

Or equivalently, using cyclic permutation. We can express the gradient equation using the parameter vector $\alpha = K_y^{-1} y$:

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \frac{1}{2} \text{Tr} \left((\alpha \alpha^T - K_y^{-1}) \frac{\partial K_y}{\partial \theta_j} \right)$$

□

B.3 Analytical Derivatives of the Matérn 5/2 Kernel

This is the mathematical derivation of the analytical derivatives of the Matérn 5/2 kernel, although the base equation of the Matérn class is well-documented in standard literature [12]. Here we present the exact continuous calculus required to prove that the Gaussian Process natively supports the arbitrage bounds (gradient for Calendar Arbitrage, Hessian for Butterfly Arbitrage) for our two-dimensional (K, T) anisotropic volatility surface, removing any reliance on finite-difference approximations.

B.3.1 The GP Linear Operator Property

Proof. Derived in the Equation B.3, we define the predictive mean of a Gaussian Process as:

$$\bar{f}_* = k_*^T (K_{yy})^{-1} y$$

However, now we optimise the multivariate calculus for the algorithm. Hence, we define $k_* = K(X, x_*)$, which is a row vector, as the cross-covariance vector between the training data and a single test coordinate x_* , and y is the vector of observed targets. Because the test coordinate x_* only appears inside the kernel function vector k_* , we can again define the pre-computed weight vector $\alpha = (K_{yy})^{-1} y$.

Because arbitrage constraints (Section 2.2) are strictly directional. We differentiate with respect to a specific dimension i of the test coordinate $x_{*,i}$ (e.g., Strike K or Maturity T):

$$\frac{\partial \bar{f}_*}{\partial x_{*,i}} = \left(\frac{\partial k_*}{\partial x_{*,i}} \right)^T \alpha$$

Similarly, the second derivative (the Hessian diagonal) required for evaluating butterfly convexity becomes:

$$\frac{\partial^2 \bar{f}_*}{\partial x_{*,i}^2} = \left(\frac{\partial^2 k_*}{\partial x_{*,i}^2} \right)^T \alpha$$

B.3.2 The Matérn 5/2 Kernel and Mahalanobis Distance

To compute the derivatives of k_* , by the chain rule, we must differentiate the kernel function itself. We define the anisotropic Matérn 5/2 kernel between a training point x and a test point x_* as:

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right) \exp(-\sqrt{5}r)$$

where r is the anisotropic Mahalanobis distance. For a two-dimensional input space of Strike ($x_1 = K$) and Maturity ($x_2 = T$) with independent length-scales ℓ_1 and ℓ_2 , the Mahalanobis distance is:

$$r(x, x_*) = \sqrt{\frac{(x_{*,1} - x_1)^2}{\ell_1^2} + \frac{(x_{*,2} - x_2)^2}{\ell_2^2}}$$

B.3.3 The Derivation: First Derivative (The Gradient)

We apply the chain rule to differentiate the kernel with respect to a specific spatial input coordinate $x_{*,i}$:

$$\frac{\partial k_{5/2}}{\partial x_{*,i}} = \frac{\partial k_{5/2}}{\partial r} \frac{\partial r}{\partial x_{*,i}}$$

Step A: Differentiate the Kernel with Respect to Distance (r)

Applying the product rule to $k_{5/2}(r)$:

$$\frac{\partial k_{5/2}}{\partial r} = \sigma_f^2 \left[\left(\sqrt{5} + \frac{10}{3}r \right) \exp(-\sqrt{5}r) - \sqrt{5} \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right) \exp(-\sqrt{5}r) \right]$$

Factoring out $\sigma_f^2 \exp(-\sqrt{5}r)$ and simplifying the polynomial:

$$\begin{aligned} \frac{\partial k_{5/2}}{\partial r} &= \sigma_f^2 \exp(-\sqrt{5}r) \left[\sqrt{5} + \frac{10}{3}r - \sqrt{5} - 5r - \frac{5\sqrt{5}}{3}r^2 \right] \\ \therefore \frac{\partial k_{5/2}}{\partial r} &= -\frac{5}{3}\sigma_f^2(1 + \sqrt{5}r^2) \exp(-\sqrt{5}r) \end{aligned}$$

Step B: Differentiate the Distance with Respect to the Coordinate ($x_{*,i}$)

Differentiating the Mahalanobis distance r using the chain rule:

$$\frac{\partial r}{\partial x_{*,i}} = \frac{1}{2r} \cdot 2 \frac{(x_{*,i} - x_i)}{\ell_i^2} = \frac{x_{*,i} - x_i}{r\ell_i^2}$$

Step C: The Complete First Derivative

Multiplying Step A and Step B (noting that the r in the denominator cancels with the r in the numerator):

$$\frac{\partial k_{5/2}}{\partial x_{*,i}} = -\frac{5}{3}\sigma_f^2(1 + \sqrt{5}r) \exp(-\sqrt{5}r) \frac{(x_{*,i} - x_i)}{\ell_i^2}$$

This first derivative evaluates the continuous local slope of the surface, natively defining the calendar arbitrage constraint when evaluated with respect to Maturity (T).

B.3.4 The Derivation: Second Derivative (The Hessian)

To evaluate butterfly convexity, we require the second partial derivative with respect to Strike (K). Let $x_{*,i} = K$. We differentiate the first derivative again using the product rule. Let the derivative of Mahalanobis distance r with respect to a single test coordinate $x_{*,i}$ be defined as $D_K = \frac{x_{*,i} - x_i}{\ell_i^2}$ for simplicity. Hence, the first derivative simplifies to:

$$\frac{\partial k_{5/2}}{\partial x_{*,i}} = -\frac{5}{3}\sigma_f^2 \left[(1 + \sqrt{5}r) \exp(-\sqrt{5}r) \right] D_K$$

By applying the product rule, we obtain:

$$\frac{\partial^2 k_{5/2}}{\partial x_{*,i}^2} = -\frac{5}{3}\sigma_f^2 \left[\frac{\partial}{\partial x_{*,i}} \left((1 + \sqrt{5}r) \exp(-\sqrt{5}r) \right) \cdot D_K + (1 + \sqrt{5}r) \exp(-\sqrt{5}r) \cdot \frac{\partial D_K}{\partial x_{*,i}} \right]$$

Step A: Inner Chain Rule

Let $u(r) = (1 + \sqrt{5}r) \exp(-\sqrt{5}r)$.

$$\frac{\partial u}{\partial r} = \sqrt{5} \exp(-\sqrt{5}r) - \sqrt{5}(1 + \sqrt{5}r) \exp(-\sqrt{5}r) = -5r \exp(-\sqrt{5}r)$$

Multiplying by $\frac{\partial r}{\partial x_{*,i}}$:

$$\frac{\partial u}{\partial x_{*,i}} = -5r \exp(-\sqrt{5}r) \frac{x_{*,i} - x_i}{r\ell_i^2} = -5 \exp(-\sqrt{5}r) D_K$$

Step B: The Second Term

The derivative of the spatial distance component D_K is simply a constant scaling factor:

$$\frac{\partial D_K}{\partial x_{*,i}} = \frac{1}{\ell_i^2}$$

Step C: The Complete Second Derivative

Substituting the components back into the product rule yields the exact analytical second derivative:

$$\frac{\partial^2 k_{5/2}}{\partial x_{*,i}^2} = -\frac{5}{3}\sigma_f^2 \left[-5 \exp(-\sqrt{5}r) D_K^2 + (1 + \sqrt{5}r) \exp(-\sqrt{5}r) \frac{1}{\ell_i^2} \right]$$

□

B.4 The Breeden-Litzenberger Density and Static Arbitrage

In this derivation, we establish the relationship between the price of a European call option and the implied risk-neutral probability density function. Although the foundational logic follows the framework established by Breeden and Litzenberger [4], we expand this by showing that the term must remain strictly positive to avoid butterfly arbitrage.

B.4.1 The Setup: Risk-Neutral Expected Payoff

Proof. Under the fundamental theorem of asset pricing, the arbitrage-free price of a European call option $C(K, T)$ with strike K and maturity T is the discounted expected value of its payoff under the risk-neutral measure $\mathbb{Q}$ (see [III. Arbitrage & Pricing Theory](#)).

Assuming a constant continuously compounded risk-free rate r , we can express the expectation as an integral over the risk-neutral probability density function (PDF) of the terminal asset price, denoted as $p(S_T)$:

$$C(K, T) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)]$$

The integral can be separated into two parts, with the first ranging from 0 to K and the second from K to ∞ . Since the payoff of the first one is zero if $S_T - K < 0$, we equate the integral to 0 and proceed with the second.

$$C(K, T) = e^{-rT} \int_K^{\infty} (S_T - K) p(S_T) dS_T \quad (\text{B.9})$$

B.4.2 The Derivation: First Derivative (The Survival Function)

To isolate the density function $p(S_T)$, we must differentiate the integral with respect to the strike price K . Because the variable of differentiation K appears in both the integrand and the lower limit of integration, we apply the Leibniz rule:

We evaluate the upper bound, $b(K) = \infty$, so $\frac{db}{dK} = 0$. The lower bound: $a(K) = K$, so $\frac{da}{dK} = 1$. And the integrand becomes $(S_T - K)p(S_T)$.

As we evaluate the boundary terms, the upper boundary term vanishes because $\frac{db}{dK} = 0$. And the lower boundary term is evaluated at $S_T = K$, yielding $(K - K)p(K) \cdot 1 = 0$. Consequently, both terms vanish, and we are left with the integral of the partial derivative of the integrand:

$$\frac{\partial C}{\partial K} = e^{-rT} \int_K^{\infty} \frac{\partial}{\partial K} [(S_T - K)p(S_T)] dS_T$$

Differentiating the inside with respect to K and simplifying:

$$\frac{\partial C}{\partial K} = e^{-rT} \int_K^\infty (-1)p(S_T)dS_T = -e^{-rT} \int_K^\infty p(S_T)dS_T \quad (\text{B.10})$$

This first derivative represents the survival function; negative discounted risk-neutral probability that the option finishes in the money (see [II. Volatility & Surface Metrics](#)).

B.4.3 The Derivation: Second Derivative (The Density)

To prove the butterfly arbitrage constraint, we must differentiate a second time with respect to K . This time, K only appears in the lower bound of the integral. Thereby, we simply apply the Fundamental Theorem of Calculus:

$$\frac{\partial^2 C}{\partial K^2} = -e^{-rT} \frac{\partial}{\partial K} \int_K^\infty p(S_T)dS_T$$

By the Fundamental Theorem of Calculus, when we differentiate an integral with respect to its lower bound, the equation simply yields the negative of the integrand evaluated at that bound:

$$\frac{\partial^2 C}{\partial K^2} = -e^{-rT} (-p(K)) = e^{-rT} p(K) \quad (\text{B.11})$$

B.4.4 The Result: The Butterfly Arbitrage Constraint

After simplification, we isolate the risk-neutral probability density $p(K)$ to obtain the Breeden-Litzenberger formula:

$$p(K) = e^{rT} \frac{\partial^2 C}{\partial K^2} \quad (\text{B.12})$$

Following Kolmogorov's foundational axioms of probability theory [25], a probability density function can never be negative. Therefore, it must hold true that $p(K)$ is strictly positive $p(K) \geq 0$. Additionally, because the discount factor e^{rT} also can never be negative, the second partial derivative of the call price must always be positive.

$$\frac{\partial^2 C}{\partial K^2} \geq 0$$

Hence, if an interpolated surface exhibits local concavity, it directly implies the existence of a negative risk-neutral probability. This violates static butterfly constraints and makes the surface financially unreliable.

□

B.5 Dupire's Equation

According to the Fundamental Theorem of Asset Pricing, for an option surface to be valid, there must exist a risk-neutral stochastic process (an SDE) that generates those prices. The foundation of our risk-neutral stochastic process model is based on standard stochastic financial theory as presented in the Shreve [20] textbook.

Furthermore, for the Dupire equation to show that this SDE exists, the local variance must be strictly positive. Hence, if a butterfly arbitrage violation occurs, the local variance becomes negative, and the diffusion term $\sigma_L S_t dW_t$ becomes mathematically undefined. Consequently, rendering the option surface invalid for financial modelling.

B.5.1 The Setup: The SDE and the FPK Equation

Proof. We assume the underlying asset S_t follows a risk-neutral stochastic differential equation (SDE):

$$dS_t = rS_t dt + \sigma_L(S_t, t)S_t dW_t \quad (\text{B.13})$$

Conceptually, the stochastic differential equation dictates that dS_t is equal to the sum of the drift term ($rS_t dt$), where r is the risk-free rate and dt is an infinitesimal step forward in time, and the diffusion term ($\sigma_L(S_t, t)S_t dW_t$). Here, $\sigma_L(S_t, t)$ represents the volatility of the asset at that particular price S_t and time t , scaled by the standard Brownian motion dW_t and the asset price S_t . Holistically, this equation represents that the change in the stock price (dS_t) is equal to the predictable interest it earned ($rS_t dt$), adjusted by a stochastic volatility diffusion term and amplified by the local volatility conditions ($\sigma_L S_t dW_t$).

Additionally, under standard stochastic calculus theory [20, 26], the transition probability density function $p(x, T)$ of the process satisfies the Fokker–Planck–Kolmogorov (FPK) forward equation:

$$\frac{\partial p}{\partial T} = \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_L^2(x, T)x^2 p) - \frac{\partial}{\partial x} (rxp) \quad (\text{B.14})$$

Where x serves as the continuous spatial dummy variable representing the state space of the terminal asset price S_T .

B.5.2 The Derivation: Differentiating the Call Price with Respect to Time

We begin with the standard risk-neutral pricing formula for a European call, expressed as an integral over the transition density $p(x, T)$. This notation was previously introduced

in Equation B.9:

$$C(K, T) = e^{-rT} \int_K^\infty (x - K) p(x, T) dx$$

To find the evolution of the price over time, we differentiate both sides with respect to maturity T using the Leibniz integral rule:

$$\frac{\partial C}{\partial T} = -re^{-rT} \int_K^\infty (x - K) p(x, T) dx + e^{-rT} \int_K^\infty (x - K) \frac{\partial p}{\partial T} dx$$

We recognise that the first term contains the original call price definition. So the equation simplifies to:

$$\frac{\partial C}{\partial T} = -rC + e^{-rT} \int_K^\infty (x - K) \frac{\partial p}{\partial T} dx \quad (\text{B.15})$$

B.5.3 Substituting the FPK Equation

In Equation B.15, the time derivative of the density, $\frac{\partial p}{\partial T}$, is located inside an integral evaluated with respect to the spatial variable x . Because T and x are independent variables, we make use of the FPK Equation defined in B.14 to map the time derivative into spatial derivatives. Substituting the FPK Forward Equation splits the integral into a diffusion term and a drift term:

$$\frac{\partial C}{\partial T} = -rC + e^{-rT} \underbrace{\int_K^\infty (x - K) \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_L^2 x^2 p) dx}_{\text{Diffusion Term}} - e^{-rT} \underbrace{\int_K^\infty (x - K) \frac{\partial}{\partial x} (rxp) dx}_{\text{Drift Term}} \quad (\text{B.16})$$

We evaluate the diffusion term using integration by parts. Let $u = (x - K)$ and $dv = \frac{\partial^2}{\partial x^2} (\sigma_L^2 x^2 p) dx$, yielding $du = dx$ and $v = \frac{\partial}{\partial x} (\sigma_L^2 x^2 p)$. Collecting the terms, we obtain:

$$\int_K^\infty (x - K) \frac{\partial^2}{\partial x^2} (\sigma_L^2 x^2 p) dx = \left[(x - K) \frac{\partial}{\partial x} (\sigma_L^2 x^2 p) \right]_K^\infty - \int_K^\infty \frac{\partial}{\partial x} (\sigma_L^2 x^2 p) dx$$

Assuming the transition density $p(x, T)$ and its first derivative decay to zero sufficiently fast as $x \rightarrow \infty$, the upper boundary evaluated at infinity vanishes. At the lower boundary $x = K$, the $(x - K)$ multiplier becomes $(K - K) = 0$. Thus, the entire first boundary term is zero. Applying the Fundamental Theorem of Calculus to the remaining integral:

$$- \int_K^\infty \frac{\partial}{\partial x} (\sigma_L^2 x^2 p) dx = - [\sigma_L^2 x^2 p]_K^\infty \quad (\text{B.17})$$

Again, evaluating at infinity yields zero due to the asymptotic decay of $p(x, T)$.

Evaluating at the lower bound K subtracts the term, neutralising the negative sign:

$$-\left(0 - \sigma_L^2(K, T)K^2p(K, T)\right) = \sigma_L^2(K, T)K^2p(K, T)$$

Next, we evaluate the drift term using integration by parts. Let $u = (x - K)$, so $du = dx$. Let $dv = \frac{\partial}{\partial x}(rxp)dx$, yielding $v = rxp$.

$$\int_K^\infty (x - K) \frac{\partial}{\partial x}(rxp)dx = [(x - K)rxp]_K^\infty - \int_K^\infty rxpdx$$

At the lower bound ($x = K$), the $(x - K)$ term resolves to zero. As $x \rightarrow \infty$, the probability density $p(x, T)$ decays to zero faster than the polynomial $(x - K)rx$ grows, causing the upper bound to vanish. We are left only with the subtracted integral:

$$- \int_K^\infty rxpdx \tag{B.18}$$

Substituting the evaluated diffusion term [B.17](#) and drift term [B.18](#) back into the time-derivative Equation [B.16](#):

$$\frac{\partial C}{\partial T} = -rC + e^{-rT} \left(\frac{1}{2} \sigma_L^2(K, T) K^2 p(K, T) \right) - e^{-rT} \left(- \int_K^\infty rxpdx \right)$$

$$\frac{\partial C}{\partial T} = -rC + \frac{1}{2} \sigma_L^2(K, T) K^2 \left(e^{-rT} p(K, T) \right) + re^{-rT} \int_K^\infty xpdx$$

B.5.4 Connecting to Breeden-Litzenberger

To isolate the local volatility term, we rewrite the remaining components strictly in terms of call price derivatives. First, applying the Breeden-Litzenberger result derived in [B.11](#) ($e^{-rT}p(K, T) = \frac{\partial^2 C}{\partial K^2}$) to the diffusion term, and expanding the integral in the drift term by adding and subtracting K :

$$re^{-rT} \int_K^\infty (x - K + K)pdx = re^{-rT} \int_K^\infty (x - K)pdx + rKe^{-rT} \int_K^\infty pdx$$

Recognizing that $e^{-rT} \int_K^\infty (x - K)pdx$ is the Call Price C , and recalling from [B.10](#) that $\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty pdx$, we substitute these identities back into the drift term:

$$r \left(C - K \frac{\partial C}{\partial K} \right)$$

Inserting these back into the main equation yields:

$$\frac{\partial C}{\partial T} = -rC + \frac{1}{2} \sigma_L^2(K, T) K^2 \frac{\partial^2 C}{\partial K^2} + rC - rK \frac{\partial C}{\partial K}$$

The $-rC$ and $+rC$ terms cancel, leaving:

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma_L^2(K, T)K^2\frac{\partial^2 C}{\partial K^2} - rK\frac{\partial C}{\partial K}$$

B.5.5 The Result: Dupire's Formula

Isolating the local variance $\sigma_L^2(K, T)$ yields Dupire's foundational local volatility formula [14]:

$$\sigma_L^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK\frac{\partial C}{\partial K}}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}} \quad (\text{B.19})$$

This formula is crucial for understanding how butterfly arbitrage violations directly translate to an economically inadmissible volatility surface. Additionally, Dupire's equation [14] converts Implied Volatility to Local Volatility for pricing exotic options (e.g., Barrier or Asian options, see [III. Arbitrage & Pricing Theory](#)). If the second derivative $\frac{\partial^2 C}{\partial K^2}$ is negative, the local variance becomes negative, rendering the local volatility an imaginary number and destabilising the pricing framework. $\square$

B.6 Mercer's Theorem and the RKHS Norm Derivation

The theoretical basis for Reproducing Kernel Hilbert Spaces (RKHS) is a cornerstone of functional analysis, originating from the eigen-decomposition theorem of Mercer [30] and formalized by Aronszajn [31]. The following proof adapts these principles to explicitly derive the RKHS norm penalty acting upon our Gaussian Process loss functional, formalizing the structural regularisation mechanism. Conceptually representing the formal penalty that constrains the Gaussian Process Regression from creating an overly jagged surface. Hence, it is necessary to formally introduce it in this paper.

B.6.1 Definition of the Kernel Operator

Proof. We begin by defining the kernel operator. Let $\mathcal{X}$ be a compact space and $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a continuous, symmetric, positive semi-definite kernel function. Then, define the integral operator $T_k : L^2(\mathcal{X}) \rightarrow L^2(\mathcal{X})$ such that:

$$(T_k f)(x) = \int_{\mathcal{X}} k(x, x') f(x') dx'$$

Since we cannot decompose the kernel directly, we have to take the continuum limit of the discrete Gram matrix $K \in \mathbb{R}^{n \times n}$, thereby obtaining the linear integral operator $T_k : L^2(\mathcal{X}) \rightarrow L^2(\mathcal{X})$. Essentially upgrading a discrete matrix K to its continuous counterpart T_k to be able to decompose the kernel by using Mercer's Theorem.

B.6.2 Mercer's Theorem (Eigen-decomposition)

By Mercer's Theorem, T_k admits an orthonormal basis of eigenfunctions $\{\phi_i\}_{i=1}^{\infty}$ and a corresponding set of non-negative eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$ ordered such that $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$. Intuitively, Mercer's Theorem guarantees that we can break our integral operator T_k into an infinite set of perfectly independent eigenfunctions and sort them by their importance (eigenvalues). So, the eigenvalue problem is defined as:

$$\int_{\mathcal{X}} k(x, x') \phi_i(x') dx' = \lambda_i \phi_i(x)$$

By Mercer's Theorem, we can express our Matern kernel as a uniformly convergent series:

$$k(x, x') = \sum_{i=1}^{\infty} \lambda_i \phi_i(x) \phi_i(x')$$

B.6.3 Construction of the RKHS ($\mathcal{H}_k$)

Having deconstructed the kernel into eigenfunctions and eigenvalues, we pick any two basis functions to construct the RKHS norm. Let $\mathcal{H}_k$ be the function space defined by the span of the eigenfunctions, formally $\mathcal{H}_k = \text{span}\{\phi_i\}_{i=1}^{\infty}$. Any function $f, g \in \mathcal{H}_k$ can be expressed as a linear combination of these basis functions:

$$f(x) = \sum_{i=1}^{\infty} c_i \phi_i(x)$$

$$g(x) = \sum_{i=1}^{\infty} d_i \phi_i(x)$$

where $c_i, d_i \in \mathbb{R}$.

B.6.4 Verification of the Reproducing Property

To prove $\mathcal{H}_k$ is a Reproducing Kernel Hilbert Space, we must show that $\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = f(x)$. Where, by definition, the inner product space for $\mathcal{H}_k$ is $\langle f, g \rangle_{\mathcal{H}_k} = \sum_{i=1}^{\infty} \frac{c_i d_i}{\lambda_i}$. Contingent on the fact that c_i, d_i must decay sufficiently fast such that the sum converges. Firstly, we express the kernel function $k(\cdot, x)$ for a fixed x in terms of the basis functions. Let $d_i = \lambda_i \phi_i(x)$:

$$k(\cdot, x) = \sum_{i=1}^{\infty} (\lambda_i \phi_i(x)) \phi_i(\cdot)$$

We substitute c_i from $f(x)$ and d_i from $k(\cdot, x)$ into the inner product definition and obtain:

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = \sum_{i=1}^{\infty} \frac{c_i (\lambda_i \phi_i(x))}{\lambda_i}$$

As a result, the λ_i terms cancel out:

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = \sum_{i=1}^{\infty} c_i \phi_i(x)$$

By the definition of $f(x)$ established priorly, this evaluates to:

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = f(x)$$

Thus, proving that $\mathcal{H}_k$ is indeed a valid RKHS.

B.6.5 Derivation of the RKHS Norm

The RKHS norm of the function $f \in \mathcal{H}_k$ can be found by its inner product with itself. Setting $g(x) = f(x)$, we imply that $d_i = c_i$:

$$\|f\|_{\mathcal{H}_k}^2 = \langle f, f \rangle_{\mathcal{H}_k} \quad (\text{B.20})$$

$$\|f\|_{\mathcal{H}_k}^2 = \sum_{i=1}^{\infty} \frac{c_i c_i}{\lambda_i}$$

$$\|f\|_{\mathcal{H}_k}^2 = \sum_{i=1}^{\infty} \frac{c_i^2}{\lambda_i}$$

It is important to note that the abstract definition of the RKHS norm in Equation B.20 cannot be evaluated computationally due to the infinite sum of spectral eigenvalues $(\sum_{i=1}^{\infty} \frac{c_i^2}{\lambda_i})$. Instead, we invoke the Representer Theorem [12, 32, 33], which guarantees that the minimiser of an infinite-dimensional RKHS functional reduces exactly to a finite matrix quadratic form:

$$\|f\|_{\mathcal{H}_k}^2 = y^T K_y^{-1} y$$

This formulation can be found in the data fit term in Equation B.5 of Appendix B.2 for the Log-Marginal Likelihood.

B.6.6 Application to the Gaussian Process Loss Functional

Under classical Tikhonov regularisation [34] (and by extension, the MAP estimation framework of Gaussian Process regression [12]), the objective functional $J(f)$ over N observations is defined as:

$$J(f) = \sum_{j=1}^N (y_j - f(x_j))^2 + \sigma_n^2 \|f\|_{\mathcal{H}_k}^2$$

Where, practically, $\sum_{j=1}^N (y_j - f(x_j))^2$ is the data error term and $\sigma_n^2 \|f\|_{\mathcal{H}_k}^2$ is the RKHS norm penalty. Finally, by substituting the derived norm into the objective functional $J(f)$, we obtain the final result:

$$J(f) = \sum_{j=1}^N \left(y_j - \sum_{i=1}^{\infty} c_i \phi_i(x_j) \right)^2 + \sigma_n^2 \sum_{i=1}^{\infty} \frac{c_i^2}{\lambda_i}$$

□

B.7 Fisher Information Matrix (FIM) for a Gaussian Process

While the general statistical definition of information matrices originates from the foundational work of Fisher [35] and standard spatial statistics literature [12], the following proof explicitly executes the second-order matrix calculus and trace expectations to extract the exact hyperparameter covariance bounds for our volatility framework.

B.7.1 The Log-Marginal Likelihood

Proof. We begin by setting up the objective function $\mathcal{L}(\theta)$ of the Log-Marginal Likelihood we defined in Section 3.3.

$$\mathcal{L}(\theta) = -\frac{1}{2}y^T K^{-1}y - \frac{1}{2}\log |K| - \frac{N}{2}\log(2\pi)$$

B.7.2 The Score Function

To find how the likelihood changes when we change a single hyperparameter (i. e., lengthscale θ_i), we take the first partial derivative to obtain the score function, which we derived in Equation B.8.

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{1}{2}y^T K^{-1} \frac{\partial K}{\partial \theta_i} K^{-1}y - \frac{1}{2}\text{Tr} \left(K^{-1} \frac{\partial K}{\partial \theta_i} \right)$$

Intuitively, the first term of the score function measures how changing the kernel affects the data fit, and the second term (the trace) measures how changing the kernel affects the RKHS norm.

B.7.3 The Second Derivative (The Hessian)

To obtain the Fisher Information Matrix, which is defined as the negative expected value of the second derivative (the Hessian matrix). Hence, we differentiate the score function, this time with respect to a second hyperparameter θ_j . For simplicity, let $K_i = \frac{\partial K}{\partial \theta_i}$ and $K_j = \frac{\partial K}{\partial \theta_j}$ to save space. We apply the product rule and obtain:

$$-\frac{1}{2} \frac{\partial}{\partial \theta_j} \text{Tr}(K^{-1} K_i) = -\frac{1}{2} \text{Tr} \left(-K^{-1} K_j K^{-1} K_i + K^{-1} K_{ij} \right)$$

Differentiating the Data Term:

$$\frac{1}{2} \frac{\partial}{\partial \theta_j} \left(y^T K^{-1} K_i K^{-1} y \right) = \frac{1}{2} y^T \left(-K^{-1} K_j K^{-1} K_i K^{-1} + K^{-1} K_{ij} K^{-1} - K^{-1} K_i K^{-1} K_j K^{-1} \right) y \quad (\text{B.21})$$

B.7.4 The Expectation

We define the Fisher Information Matrix $\mathcal{I}_{ij}$ as the negative Expectation over all possible market observations y :

$$\mathcal{I}_{ij} = -\mathbb{E}_y \left[\frac{\partial^2 \mathcal{L}}{\partial \theta_i \partial \theta_j} \right]$$

In order to evaluate the expectation of the massive quadratic form from Equation B.21. We use a trace trick from linear algebra. For any matrix A , the expectation of a quadratic form is $\mathbb{E}[y^T A y] = \text{Tr}(A \cdot \mathbb{E}[y y^T])$. Because $\mathbb{E}[y y^T]$ is simply the covariance matrix K , the rule simplifies to:

$$\mathbb{E}[y^T A y] = \text{Tr}(AK)$$

After applying the trace trick and simplifying, we are left with:

$$-\mathbb{E}[\text{Data Term}] = -\frac{1}{2} \text{Tr} \left(\left(-2K^{-1} K_j K^{-1} K_i K^{-1} + K^{-1} K_{ij} K^{-1} \right) K \right)$$

As we distribute the coefficient and matrix K across both terms, we can safely cancel out K^{-1} at the end of the brackets with the K from the trace trick:

$$-\mathbb{E}[\text{Data Term}] = \text{Tr}(K^{-1} K_j K^{-1} K_i) - \frac{1}{2} \text{Tr}(K^{-1} K_{ij})$$

B.7.5 The Result: Fisher Information Matrix

After simplifying, we now apply the negative expectation to the trace term. Because the trace contains no y variables, meaning it is entirely deterministic, the expectation does nothing. Hence, we simply invert the sign:

$$-\mathbb{E}[\text{Trace Term}] = -\frac{1}{2} \text{Tr}(K^{-1} K_j K^{-1} K_i) + \frac{1}{2} \text{Tr}(K^{-1} K_{ij})$$

Finally, we collect the terms and cancel out the second cross-derivatives ($\frac{1}{2} \text{Tr}(K^{-1} K_{ij})$) to obtain the Fisher Information matrix element $\mathcal{I}_{ij}$:

$$\mathcal{I}_{ij} = \left[\text{Tr}(K^{-1} K_j K^{-1} K_i) - \frac{1}{2} \text{Tr}(K^{-1} K_{ij}) \right] + \left[-\frac{1}{2} \text{Tr}(K^{-1} K_j K^{-1} K_i) + \frac{1}{2} \text{Tr}(K^{-1} K_{ij}) \right]$$

$$\mathcal{I}_{ij} = \frac{1}{2} \text{Tr} \left(K^{-1} \frac{\partial K}{\partial \theta_i} K^{-1} \frac{\partial K}{\partial \theta_j} \right)$$

Thereby, mathematically proving that the curvature of our optimisation landscape is entirely dependent on how the covariance matrix K reacts to changes in its hyperparameters ($\frac{\partial K}{\partial \theta}$). Because our Matérn kernel is fully differentiable, this FIM is guaranteed to exist and be well-behaved, ensuring that our L-BFGS-B optimiser will reliably converge to the safest possible volatility surface without numerical divergence.

□

Appendix C

Comprehensive Empirical Results

This appendix contains the aggregated out-of-sample Root Mean Square Error (RMSE) and discrete arbitrage violation counts for the evaluated market regimes.

Date	Method	OOS RMSE (IV)	Calendar Viol.	Butterfly Viol.
2020-03-03	Point-Estimated GP	0.051027	0	8
	Bayesian HMC GP	0.048889	2	0
2020-03-13	Point-Estimated GP	0.027621	8	321
	Bayesian HMC GP	0.145624	34	146
2020-03-17	Point-Estimated GP	0.080205	0	40
	Bayesian HMC GP	0.016872	1	40
2020-03-19	Point-Estimated GP	0.012353	223	127
	Bayesian HMC GP	0.012368	0	3
2020-03-23	Point-Estimated GP	0.049056	0	0
	Bayesian HMC GP	0.006478	0	0
2020-03-25	Point-Estimated GP	0.103873	0	0
	Bayesian HMC GP	0.026344	1	0
2022-09-16	Point-Estimated GP	0.046912	0	0
	Bayesian HMC GP	0.034505	0	0
2022-09-21	Point-Estimated GP	0.023287	0	0
	Bayesian HMC GP	0.001327	0	0
2022-09-26	Point-Estimated GP	0.007120	0	0
	Bayesian HMC GP	0.000643	0	0
2022-09-30	Point-Estimated GP	0.015139	0	0
	Bayesian HMC GP	0.005231	0	0
2022-10-05	Point-Estimated GP	0.002564	0	101
	Bayesian HMC GP	0.003105	0	10
2023-06-07	Point-Estimated GP	0.001656	0	80
	Bayesian HMC GP	0.001662	0	4
2023-06-13	Point-Estimated GP	0.022153	0	77
	Bayesian HMC GP	0.024853	0	0
2023-06-21	Point-Estimated GP	0.009758	450	113
	Bayesian HMC GP	0.006871	37	0
2023-06-26	Point-Estimated GP	0.004627	0	52
	Bayesian HMC GP	0.000791	2	0
2023-06-30	Point-Estimated GP	0.042433	0	1
	Bayesian HMC GP	0.048699	0	1
2024-01-18	Point-Estimated GP	0.068329	0	0
	Bayesian HMC GP	0.007849	0	0
2024-01-24	Point-Estimated GP	0.000330	0	2
	Bayesian HMC GP	0.000353	0	0
2024-01-30	Point-Estimated GP	0.000446	0	2
	Bayesian HMC GP	0.000444	0	2
2024-02-09	Point-Estimated GP	0.050451	0	0
	Bayesian HMC GP	0.049075	0	0
2024-02-15	Point-Estimated GP	0.000925	0	17
	Bayesian HMC GP	0.000929	0	8

Table C.1: Master Comparison of Out-of-Sample RMSE and Arbitrage Violations: Point-Estimated GP vs. Bayesian HMC GP.

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